LOG-CONCAVITY OF INVERSE Z-POLYNOMIALS OF DOWLING MATROIDS WITH ARBITRARY JOINTS

KNOWLEDGE SENSE

Abstract. Let G be a finite group of order m, let $n \geq 1$, and let $X \subseteq \{1, \ldots, n\}$. Consider the frame matroid $\mathsf{Q}_n(G; X)$ of the gain graph on n vertices that has one link with each gain $g \in G$ between any two vertices and a half edge at each vertex of X. This family contains the Dowling geometries, the jointless Dowling geometries, the matroids of the reflection arrangements of types B_n and D_n, the braid matroids, and the cycle matroids of complete graphs with an apex vertex joined to an arbitrary set of vertices. We prove that the inverse Z-polynomial of $\mathsf{Q}_n(G; X)$ depends only on n, m and $|X|$, that its coefficients are positive, and that they form a strictly log-concave sequence. In particular, the conjecture of Gao, Ruan and Xie on the log-concavity of inverse Z-polynomials holds for all these matroids, for every finite group G. The proof combines the deletion formula of Braden, Ferroni, Matherne and Nepal with a closed formula for the exponential generating function of the inverse Z-polynomials of Dowling geometries, which we derive from the generating function of Ferroni and Larson. Each coefficient sequence is then written as a positive mixture of products of two log-concave sequences with explicit weights, and log-concavity of the weights is transferred to the coefficients through the discrete Prékopa–Leindler inequality of Klartag and Lehec.

1. Introduction

Let M be a loopless matroid with lattice of flats $\mathcal{L}(M)$. The Kazhdan–Lusztig polynomial $P_M(t)$ of Elias, Proudfoot and Wakefield [4] and the Z-polynomial $Z_M(t)$ of Proudfoot, Xu and Young [16] are characterized by the conditions $P_M = 1$ if $\operatorname{rk} M = 0$, $\deg P_M < \frac{1}{2} \operatorname{rk} M$ if $\operatorname{rk} M > 0$, and the palindromicity of

$$Z_M(t) = \sum_{F \in \mathcal{L}(M)} t^{\operatorname{rk} F} P_{M/F}(t)$$

in degree $\operatorname{rk} M$ [16, 2]. Inverting P and Z in the incidence algebra of $\mathcal{L}(M)$ produces the inverse Kazhdan–Lusztig polynomial $Q_M(t)$ of Gao and Xie [11] and the inverse Z-polynomial $Y_M(t)$, which was introduced by Ferroni,

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Matherne, Stevens and Vecchi [7] and studied by Gao, Ruan and Xie [10]; we recall the definition in Section 2. Gao, Ruan and Xie proved that $Y_M(t)$ has nonnegative coefficients and is multiplicative under direct sums [10, Propositions 2.1 and 2.3], and recorded that it is palindromic of degree $\operatorname{rk} M$ [10, Lemma 2.2]; they computed it for uniform and sparse paving matroids and proposed the following conjecture.

Conjecture 1.1 ([10, Conjecture 1.5])**.** *For every matroid M, the coefficients of $Y_M(t)$ form a log-concave sequence with no internal zeros.*

They proved Conjecture 1.1 for sparse paving matroids [10, Theorem 1.6]. Braden, Ferroni, Matherne and Nepal [1] later established a deletion formula for $Y_M(t)$ (Theorem 2.3 below) and listed Conjecture 1.1 among the main open problems on these invariants [1, Conjecture 1.2(d)]. An equivariant version of Y_M has been considered in [8].

In this paper we prove Conjecture 1.1, with strict inequalities, for a family of matroids built from finite groups. Let G be a finite group of order m, let $n \geq 1$, and let $X \subseteq [n] = \{1, \ldots, n\}$. Consider the gain graph on the vertex set $[n]$ that has, for every pair of distinct vertices, one link carrying each gain $g \in G$, together with a half edge (a *joint*) at every vertex of X. Its frame matroid is denoted $\mathsf{Q}_n(G; X)$, and we call it the *Dowling matroid with joints X*; see Section 3 for the precise definition. For $X = [n]$ we obtain the Dowling geometry $\mathsf{Q}_n(G)$ of Dowling [3], and for $X = \emptyset$ the jointless Dowling geometry. The whole family was considered by Zaslavsky [19, Section 5.4, Example 2], following the terminology of Kahn and Kung [13]. For $G = \mathbb{Z}/2\mathbb{Z}$, the cases $X = [n]$ and $X = \emptyset$ are the matroids of the reflection arrangements of types B_n and D_n. For the trivial group, $\mathsf{Q}_n(1; X)$ is the cycle matroid of the graph obtained from the complete graph on $\{0, 1, \ldots, n\}$ by deleting the edges $0i$ with $i \notin X$; in particular $\mathsf{Q}_n(1)$ is the braid matroid K_{n+1}.

Theorem 1.2. *Let G be a finite group of order $m \geq 1$, let $n \geq 1$, let $X \subseteq [n]$ and $k = |X|$. Let r be the rank of $\mathsf{Q}_n(G; X)$ and write $Y_{\mathsf{Q}_n(G;X)}(t) = \sum_{p=0}^{r} y_p t^p$.*

(i) *$r = n - 1$ if $k = 0$ and either $m = 1$ or $n = 1$, and $r = n$ otherwise. If $m = 1$ and $k = 0$, then $\mathsf{Q}_n(G; \emptyset)$ is isomorphic to the braid matroid K_n.*

(ii) *$Y_{\mathsf{Q}_n(G;X)}$ depends only on n, m and k. Its coefficients satisfy $y_p = y_{r-p} > 0$ for $0 \leq p \leq r$.*

(iii) *$y_p^2 > y_{p-1} y_{p+1}$ for $1 \leq p \leq r - 1$.*

(iv) *If $n \geq 2$ and $r = n$, then*

$$y_0 = y_n = \big((m-1)(n-1) + k\big) \prod_{h=1}^{n-2} (mh + 1).$$

For $n = 1$ the matroid is empty if $k = 0$ and consists of a single coloop if $k = 1$, so that $Y = 1$ or $Y = 1 + t$. No commutativity is assumed in

Theorem 1.2: a nonabelian group has the same inverse Z-polynomials as a cyclic group of the same order. When $r \geq 4$, three links forming a triangle with gain product 1 are a circuit of size $3 < r$, so $\mathsf{Q}_n(G;X)$ is not paving; Theorem 1.2 therefore overlaps [10, Theorem 1.6] only in low rank.

Corollary 1.3. *Conjecture* 1.1 *holds, with strict log-concavity, for the Dowling geometries $\mathsf{Q}_n(G)$ of all finite groups G, for the braid matroids K_n, for the matroids of the reflection arrangements of types B_n and D_n, and for the cycle matroids of the graphs obtained from a complete graph by deleting some of the edges at one vertex.*

A central ingredient is a closed form for the inverse Z-polynomials of Dowling geometries. For a real number β and an integer l put

$$c_l(\beta) = \frac{\beta(\beta+1)\cdots(\beta+l-1)}{l!} \quad (l \geq 0), \qquad c_l(\beta) = 0 \quad (l < 0).$$

Theorem 1.4. *Let G be a finite group of order m and put $a = 1/m$. Let $v = v(x,t)$ be the formal power series in x with coefficients in $\mathbb{Q}[t]$ determined by $v(0,t) = 0$ and $v\,e^{-tv^2/2} = mx$. Then*

$$\sum_{n\geq 0} Y_{\mathsf{Q}_n(G)}(t)\,\frac{x^n}{n!} = \frac{e^{atv^2/2}}{(1-v)^a(1-tv)^a}.$$

Equivalently, for $n \geq 0$ and $0 \leq p \leq n$, with $\lambda = (n+a)/2$,

$$[t^p]\,Y_{\mathsf{Q}_n(G)}(t) = n!\,m^n \sum_{j=0}^{\min(p,n-p)} \frac{(\lambda-j)\lambda^{j-1}}{j!}\,c_{p-j}(a)\,c_{n-p-j}(a),$$

where the factor $(\lambda-j)\lambda^{j-1}/j!$ equals 1 for $j = 0$.

For the trivial group this gives $\sum_{n\geq 0} Y_{\mathsf{K}_{n+1}}(t)x^n/n! = e^{tv^2/2}(1-v)^{-1}(1-tv)^{-1}$ with $ve^{-tv^2/2} = x$.

Outline of the proof. Theorem 1.4 follows from the generating function for Z-polynomials of Dowling geometries obtained by Ferroni and Larson [5, 6]: the relation $Z \cdot Z^{-1} = 1$ in the incidence algebra, with Z^{-1} expressed through Y as in (2.1), becomes a functional equation, and the two nested compositional inverses in their formula cancel after the change of variables $x = g_t(u)/m$ of Lemma 4.3 (Section 4). By the deletion formula of [1], removing a joint changes Y by an amount that, for $m \geq 2$, does not depend on the remaining joints; for $m = 1$ it depends on their number, and the formula applies as long as at least two joints are present, since the last joint is a coloop (Proposition 3.6). For $m \geq 2$ it follows that $Y_{\mathsf{Q}_n(G;X)}$ is an affine function of $k = |X|$, interpolating between the Dowling geometry and the jointless Dowling geometry; for $m = 1$ we obtain an expansion with nonnegative coefficients in terms of braid matroids.

The coefficient formula in Theorem 1.4 does not exhibit log-concavity, because for $0 < a < 1$ the sequence $c_l(a)$ is log-convex. An Euler-operator

identity (Lemma 2.4) raises the parameter from a to $a+1$ without changing the total degree. In every case of Theorem 1.2 with $r = n \geq 2$ we obtain a representation

$$y_p = C \sum_{j=0}^{\min(p,n-p)} W_j \, d_{p-j} \, d_{n-p-j}, \qquad d_l = c_l(a+1), \tag{1.1}$$

with a constant $C > 0$ and explicit weights $W_0, \ldots, W_{\lfloor n/2 \rfloor}$. The sequence (d_l) has strictly decreasing ratios, and Lemma 2.8, which rests on the discrete Prékopa–Leindler inequality of Klartag and Lehec [14], shows that (1.1) is strictly log-concave whenever the weights are positive and log-concave. Most of the work (Sections 5 and 6) consists in proving that the weights have these properties for all n, m and k. We compare W_j with the weights U_j of the Dowling geometry, which are explicit and strictly log-concave. In all cases but one we show that the normalized second difference $(q_{j-1}+q_{j+1}-2q_j)/q_j$ of $q_j = W_j/U_j$ is smaller than $\log\bigl(U_j^2/(U_{j-1}U_{j+1})\bigr)$; in the remaining case $m = 2$, $j = 1$ we compare the products q_0q_2 and q_1^2 directly, using explicit bounds on q_0, q_1, q_2.

Dependencies. Besides standard facts on Kazhdan–Lusztig invariants of matroids, the proof uses the following results, which we use without proof: the deletion formula for inverse Z-polynomials [1, Theorem 1.4]; the generating function for the Z-polynomials of braid matroids [5, Propositions 2.3 and 2.8, Corollary 3.2] and of Dowling geometries [6, Theorem 1.1 and Proposition 3.2]; the leading coefficient of the Kazhdan–Lusztig polynomial of an odd-rank Dowling geometry [6, Example 4.2]; and the inequality of Klartag and Lehec [14] in the form of Theorem 3 of [12] (numbered as in arXiv:1905.04038v1). The flats, restrictions and contractions of $\mathsf{Q}_n(G;X)$ are derived directly from the rank function of frame matroids [18, 19].

Organization. Section 2 collects facts on Kazhdan–Lusztig invariants, formal power series and log-concavity, and proves the transfer lemma. Section 3 describes the flats of $\mathsf{Q}_n(G;X)$ and computes the effect of deleting a joint. Section 4 proves Theorem 1.4 and the case $X = [n]$ of Theorem 1.2. Section 5 treats nontrivial groups and Section 6 the trivial group. Section 7 completes the proof of Theorem 1.2 and lists some examples.

2. Preliminaries

2.1. **Kazhdan–Lusztig invariants.** We follow [15] for matroid terminology. For a loopless matroid M on E we write $\mathcal{L}(M)$ for its lattice of flats and rk for its rank function. For flats $F \subseteq F'$, the matroid $M|F'/F$ is loopless and its lattice of flats is the interval $[F, F']$ of $\mathcal{L}(M)$. The simplification $\mathrm{si}(M)$ has the same lattice of flats as M.

Theorem 2.1 ([16, 2]; see [1, Theorem 2.1])**.** *There is a unique way to assign to every loopless matroid M polynomials $P_M(t), Z_M(t) \in \mathbb{Z}[t]$ such*

that $P_M = 1$ *if* $\operatorname{rk} M = 0$, $\deg P_M < \frac{1}{2}\operatorname{rk} M$ *if* $\operatorname{rk} M > 0$, *and* $Z_M(t) = \sum_{F\in\mathcal{L}(M)} t^{\operatorname{rk} F} P_{M/F}(t)$ *is palindromic of degree* $\operatorname{rk} M$.

For a loopless matroid M we put $\tau(M) = [t^{(\operatorname{rk} M-1)/2}]P_M(t)$ if $\operatorname{rk} M$ is odd and $\tau(M) = 0$ if $\operatorname{rk} M$ is even.

Let $\mathcal{I}$ be the incidence algebra of $\mathcal{L}(M)$ over $\mathbb{Z}[t, t^{-1}]$, whose elements assign an element $\alpha_{FF'}$ to every pair of flats $F \subseteq F'$, with product $(\alpha\beta)_{FF'} = \sum_{F\subseteq H\subseteq F'} \alpha_{FH}\beta_{HF'}$. The element $Z \in \mathcal{I}$ with $Z_{FF'} = Z_{M|F'/F}(t)$ satisfies $Z_{FF} = 1$, so it is invertible. Following [1, Section 2.2] (see also [10, (2)–(3)]), the *inverse* Z*-polynomial* $Y_M(t)$ is defined by

$$(Z^{-1})_{FF'} = (-1)^{\operatorname{rk} F'-\operatorname{rk} F} Y_{M|F'/F}(t) \qquad (F \subseteq F'). \tag{2.1}$$

For a matroid with loops one puts $Y_M = Y_{M\setminus\text{loops}}$. Evaluating $Z \cdot Z^{-1} = 1$ at the pair $(\emptyset, E)$ gives, for every loopless M of positive rank,

$$\sum_{F\in\mathcal{L}(M)} (-1)^{\operatorname{rk} M-\operatorname{rk} F} Z_{M|F}(t)\, Y_{M/F}(t) = 0. \tag{2.2}$$

Lemma 2.2. *Let* M, M_1, M_2 *be loopless matroids.*

(a) P_M, Z_M, Y_M *and* $\tau(M)$ *depend only on the lattice* $\mathcal{L}(M)$*; in particular they are unchanged under simplification.*
(b) $P_{M_1\oplus M_2} = P_{M_1}P_{M_2}$, $Z_{M_1\oplus M_2} = Z_{M_1}Z_{M_2}$ *and* $Y_{M_1\oplus M_2} = Y_{M_1}Y_{M_2}$.
(c) *If* $M = M_1 \oplus \cdots \oplus M_s$ *and at least two of the* M_i *have positive rank, then* $\tau(M) = 0$.
(d) $Y_M = 1$ *if* $\operatorname{rk} M = 0$, *and* $Y_M = 1+t$ *if* M *consists of a single coloop.*

Proof. (a) The conditions in Theorem 2.1 and in (2.1) involve only the lattice of flats, its rank function and its intervals; by induction on the rank, all four invariants are determined by the ranked lattice.

(b) The flats of $M = M_1 \oplus M_2$ are the pairs (F_1, F_2) of flats, and $M/(F_1 \cup F_2) = M_1/F_1 \oplus M_2/F_2$. We prove the first identity by induction on $\operatorname{rk} M$, the case of rank 0 being trivial. Using the induction hypothesis for the contractions by nonempty flats,

$$Z_M - Z_{M_1}Z_{M_2} = P_M - P_{M_1}P_{M_2}.$$

The left side is palindromic of degree $\operatorname{rk} M$, and the right side has degree less than $\frac{1}{2}\operatorname{rk} M$; hence both vanish. The identity for Z follows from its definition, and the identity for Y is [10, Proposition 2.3].

(c) Let $r_i = \operatorname{rk} M_i$ and $r = \sum_i r_i$. We may assume that r is odd. By (b), $\deg P_M \le \sum_{r_i>0} \frac{r_i-1}{2} \le \frac{r-2}{2} < \frac{r-1}{2}$, since at least two r_i are positive. Hence $\tau(M) = 0$.

(d) The first statement is the definition. If M consists of a single coloop, then $\mathcal{L}(M) = \{\emptyset, E\}$, $Z_M = 1 + t$, and (2.2) gives $Z_M - Y_M = 0$. □

Theorem 2.3 ([1, Theorem 1.4])**.** *Let M be a loopless matroid and let e be an element of M that is not a coloop. Then*

$$Y_M(t) = Y_{M\setminus e}(t) + (1+t)\,Y_{M/e}(t) - \sum_{F\in\mathcal{T}_e} \tau\big(M|F/e\big)\,t^{\operatorname{rk} F/2}\,Y_{M/F}(t),$$

where $\mathcal{T}_e = \{F \in \mathcal{L}(M) : e \in F \text{ and } F\setminus\{e\} \notin \mathcal{L}(M)\}$.

Since $\tau(M|F/e) = 0$ unless $\operatorname{rk} F$ is even, all exponents of t are integers.

2.2. **Formal power series.** All power series are formal. For a series f with constant term 1 and $\beta \in \mathbb{R}$ we put $f^\beta = \sum_{i\ge0}\binom{\beta}{i}(f-1)^i$; then $(fg)^\beta = f^\beta g^\beta$ and $(f^\beta)^{\beta'} = f^{\beta\beta'}$, and composition with a series without constant term commutes with these operations. We use the following form of the Lagrange inversion formula; see [17, Section 5.4]. Let K be a $\mathbb{Q}$-algebra and $\varphi \in K[[u]]$ with $\varphi(0) = 1$. Then there is a unique $v \in K[[z]]$ with $v(0) = 0$ and $v = z\varphi(v)$, and for every $F \in K[[u]]$ and $n \ge 0$,

$$[z^n]F(v) = [u^n]\,F(u)\,\varphi(u)^n\Big(1 - \frac{u\varphi'(u)}{\varphi(u)}\Big).$$

Indeed, $[z^n]F(v)$ is the formal residue of $F(v(z))z^{-n-1}$, and the substitution $z = u/\varphi(u)$ transforms this residue into that of $F(u)\varphi(u)^{n+1}u^{-n-1}\,\frac{d}{du}\frac{u}{\varphi(u)}$. We only need $K = \mathbb{Q}[t]$ and $\varphi(u) = e^{tu^2/2}$:

$$v = z\,e^{tv^2/2} \quad\Longrightarrow\quad [z^n]F(v) = [u^n]\,F(u)\,(1-tu^2)\,e^{ntu^2/2}. \tag{2.3}$$

We also use series in two variables ξ, η. Put

$$z = \xi\eta, \qquad R = (1-\xi)(1-\eta),$$

and for a series $\Phi \in \mathbb{R}[[\xi,\eta]]$ let $[\Phi]_N$ be its homogeneous component of degree N. Since $R^{-\beta} = \sum_{\alpha,\gamma\ge0} c_\alpha(\beta)c_\gamma(\beta)\xi^\alpha\eta^\gamma$,

$$[\xi^{N-p}\eta^p]\,z^jR^{-\beta} = c_{N-p-j}(\beta)\,c_{p-j}(\beta). \tag{2.4}$$

For $\Phi \in \mathbb{R}[[\xi,\eta]]$ the substitution $\xi = u$, $\eta = tu$ gives a series $\Phi(u,tu) \in \mathbb{R}[t][[u]]$, and

$$[u^Nt^p]\,\Phi(u,tu) = [\xi^{N-p}\eta^p]\,\Phi. \tag{2.5}$$

Under this substitution $z = tu^2$. For $\theta \in \mathbb{R}$ and an integer j let

$$w_j(\theta) = [z^j]\,(1-z)e^{\theta z} = \frac{(\theta-j)\,\theta^{j-1}}{j!} \quad (j\ge0), \qquad w_j(\theta) = 0 \quad (j<0); \tag{2.6}$$

for $j = 0$ the middle expression equals 1.

Lemma 2.4. *Let $N \ge 0$ and $\beta, \theta \in \mathbb{R}$.*

(a) *If $j \ge 0$ and $N + \beta - 2j \ne 0$, then*

$$[z^jR^{-\beta}]_N = \frac{\beta}{N+\beta-2j}\,[z^j(1-z)R^{-\beta-1}]_N.$$

(b) *If* $2\theta = N + \beta \neq 0$, *then*

$$[(1-z)e^{\theta z}R^{-\beta}]_N = \frac{\beta}{N+\beta}\,[(1-z)e^{\theta z}R^{-\beta-1}]_N.$$

Proof. Let $\mathcal{E} = \xi\partial_\xi + \eta\partial_\eta$; it acts on homogeneous components of degree N as multiplication by N. We have $\mathcal{E}z = 2z$ and $\mathcal{E}R = -\xi - \eta + 2\xi\eta = R - (1-z)$, hence

$$(\mathcal{E}+\beta)R^{-\beta} = \beta(1-z)R^{-\beta-1}.$$

Consequently $(\mathcal{E}+\beta-2j)(z^jR^{-\beta}) = \beta z^j(1-z)R^{-\beta-1}$, and taking components of degree N gives (a). Similarly

$$(\mathcal{E}+\beta)(e^{\theta z}R^{-\beta}) = 2\theta z\, e^{\theta z}R^{-\beta} + \beta(1-z)e^{\theta z}R^{-\beta-1}.$$

Taking components of degree N and using $2\theta = N+\beta$, the left side minus the first term on the right becomes $(N+\beta)[(1-z)e^{\theta z}R^{-\beta}]_N$, which gives (b). □

Corollary 2.5. *Let* $n \geq 0$, $\beta > 0$, $J = \lfloor n/2 \rfloor$, *and let* $V(z) = \sum_j v_j z^j \in \mathbb{R}[[z]]$. *Suppose* $n + \beta - 2j > 0$ *for* $0 \leq j \leq J$, *and put* $v_{-1} = 0$ *and*

$$W_j = \beta\Big(\frac{v_j}{n+\beta-2j} - \frac{v_{j-1}}{n+\beta-2j+2}\Big) \qquad (0 \leq j \leq J).$$

Then for $0 \leq p \leq n$,

$$[\xi^{n-p}\eta^p]\,V(z)R^{-\beta} = \sum_{j=0}^{\min(p,n-p)} W_j\, c_{p-j}(\beta+1)\, c_{n-p-j}(\beta+1).$$

Proof. Only the terms with $j \leq J$ contribute to the component of degree n. By Lemma 2.4(a),

$$[V(z)R^{-\beta}]_n = \sum_{j=0}^{J} \frac{\beta v_j}{n+\beta-2j}\,[z^jR^{-\beta-1} - z^{j+1}R^{-\beta-1}]_n,$$

and the term $z^{J+1}R^{-\beta-1}$ has no component of degree n. Collecting the coefficient of $z^jR^{-\beta-1}$ and applying (2.4) gives the claim. □

2.3. **Log-concavity.** A sequence of positive numbers $(e_i)_{i\in I}$ indexed by an interval I of integers has *nonincreasing ratios* if e_{i+1}/e_i is nonincreasing in i; equivalently $e_i^2 \geq e_{i-1}e_{i+1}$ whenever $i \pm 1 \in I$. A finite sequence $(e_0, \ldots, e_N)$ of positive numbers is *strictly log-concave* if $e_i^2 > e_{i-1}e_{i+1}$ for $0 < i < N$.

Lemma 2.6. *Let* $(e_i)_{i\in I}$ *be positive with nonincreasing ratios, let* $i, i' \in I$, *and put* $u = \lfloor (i+i')/2 \rfloor$ *and* $u' = \lceil (i+i')/2 \rceil$. *Then* $e_ie_{i'} \leq e_ue_{u'}$. *If the ratios are strictly decreasing and* $|i - i'| \geq 2$, *the inequality is strict.*

Proof. Let $i \leq i'$. If $i' - i \geq 2$, then $e_{i+1}e_{i'-1}/(e_ie_{i'}) = (e_{i+1}/e_i)/(e_{i'}/e_{i'-1}) \geq 1$, with strict inequality if the ratios are strictly decreasing. Repeating this step until the indices differ by at most one proves the lemma. □

Theorem 2.7 (Klartag–Lehec [14]; see [12, arXiv:1905.04038v1, Theorem 3]). *Let $f, g, h, k\colon \mathbb{Z} \to [0,\infty)$ satisfy*

$$f(x)\,g(y) \le h\Big(\Big\lfloor \frac{x+y}{2} \Big\rfloor\Big)\, k\Big(\Big\lceil \frac{x+y}{2} \Big\rceil\Big) \qquad \text{for all } x, y \in \mathbb{Z}.$$

Then $\big(\sum_x f(x)\big)\big(\sum_y g(y)\big) \le \big(\sum_x h(x)\big)\big(\sum_y k(y)\big)$.

Lemma 2.8. *Let $n \ge 2$, $J = \lfloor n/2 \rfloor$ and $\beta > 0$. Let $W_0, \dots, W_J$ be positive numbers with $W_j^2 \ge W_{j-1} W_{j+1}$ for $1 \le j \le J-1$, put $d_l = c_l(\beta+1)$ and*

$$S_p = \sum_{j=0}^{\min(p,n-p)} W_j\, d_{p-j}\, d_{n-p-j} \qquad (0 \le p \le n).$$

Then $S_p = S_{n-p} > 0$ for $0 \le p \le n$, and $S_p^2 > S_{p-1} S_{p+1}$ for $1 \le p \le n-1$.

Proof. The sequence d_l is positive for $l \ge 0$, and $d_{l+1}/d_l = 1 + \beta/(l+1)$ is strictly decreasing. Symmetry is clear, and the term $j = 0$ shows $S_p > 0$. Extend W by $W_j = 0$ for $j \notin [0, J]$ and $d_l = 0$ for $l < 0$. Fix $1 \le p \le n-1$ and define functions on $\mathbb{Z}$ by

$$f(i) = W_i\, d_{p-1-i}\, d_{n-p+1-i}, \qquad g(i) = W_i\, d_{p+1-i}\, d_{n-p-1-i},$$
$$h(i) = W_i\, d_{p-i}\, d_{n-p-i},$$

so that $\sum f = S_{p-1}$, $\sum g = S_{p+1}$ and $\sum h = S_p$. Let $i, i' \in \mathbb{Z}$ with $f(i)g(i') > 0$ and put $u = \lfloor (i+i')/2 \rfloor$, $u' = \lceil (i+i')/2 \rceil$. Then $i, i' \in [0, J]$, and the indices $p-1-i$, $n-p+1-i$, $p+1-i'$, $n-p-1-i'$ are nonnegative. Hence $u, u' \in [0, J]$; moreover $\{p-u, p-u'\}$ is the balanced pair of integers with the same sum as $\{p-1-i,\, p+1-i'\}$, and $\{n-p-u, n-p-u'\}$ is the balanced pair with the same sum as $\{n-p+1-i,\, n-p-1-i'\}$. All these indices are nonnegative. Applying Lemma 2.6 to W on $[0, J]$ and twice to d on $[0,\infty)$ gives $f(i)g(i') \le h(u)h(u')$. The differences within the two pairs of d-indices are $i' - i - 2$ and $i' - i + 2$, which cannot both have absolute value at most 1; so one of the two applications to d is strict, and

$$h(u)\, h(u') > f(i)\, g(i') \qquad \text{whenever } f(i)g(i') > 0.$$

The functions f, g have finite support, and $f(0)g(0) > 0$. Hence

$$\rho = \min\Big\{ \frac{h(\lfloor \frac{i+i'}{2} \rfloor)\, h(\lceil \frac{i+i'}{2} \rceil)}{f(i)g(i')} : f(i)g(i') > 0 \Big\}$$

exists and $\rho > 1$. The functions ρf, g, h, h satisfy the hypothesis of Theorem 2.7, which gives $\rho\, S_{p-1} S_{p+1} \le S_p^2$. Since $S_{p\pm1} > 0$ and $\rho > 1$, we obtain $S_p^2 > S_{p-1} S_{p+1}$. □

We shall use the following elementary consequence of the arithmetic–geometric mean inequality several times. Let $q_{j-1}, q_j, q_{j+1} > 0$ and $\delta = q_{j-1} + q_{j+1} - 2q_j$. Then

$$\log \frac{q_{j-1} q_{j+1}}{q_j^2} \le 2 \log\Big(1 + \frac{\delta}{2q_j}\Big) \quad \text{if } \delta > -2q_j, \tag{2.7}$$

so that $q_{j-1}q_{j+1} \le q_j^2$ if $\delta \le 0$, and $\log(q_{j-1}q_{j+1}/q_j^2) < \delta/q_j$ if $\delta > 0$.

3. Gain graphs and the matroids $\mathsf{Q}_n(G;X)$

3.1. **Definition.** Let G be a finite group with identity 1 and order m, let $n \ge 0$ and $X \subseteq [n]$. Let $E = E_n(G;X)$ be the set consisting of the *links* $e_{ij}(g)$ for $1 \le i < j \le n$ and $g \in G$, and the *joints* h_i for $i \in X$. The link $e_{ij}(g)$ joins i and j; it has gain g when traversed from i to j and gain g^{-1} when traversed from j to i. The joint h_i is a half edge at i. For $S \subseteq E$ we consider the graph $([n], S)$. A *circle* is a closed path of links without repeated vertices; it may have length two. A circle is *balanced* if the product of the gains of its links, in the order of traversal, is 1; this does not depend on the starting point or the direction. A connected component of $([n], S)$ is *balanced* if it contains no joint and all its circles are balanced. Let $b(S)$ be the number of balanced components, isolated vertices included, and

$$\mathrm{rk}(S) = n - b(S). \tag{3.1}$$

This is the rank function of the frame matroid of the gain graph $([n], E)$ [18, Section 2], [19, Section 1]. We denote this matroid by $\mathsf{Q}_n(G;X)$ and put $\mathsf{Q}_n(G) = \mathsf{Q}_n(G;[n])$. In the notation of [19], $\mathsf{Q}_n(G;X)$ is the frame matroid of the partially filled group expansion $G \cdot K_n^{(X)}$, and $\mathsf{Q}_n(G)$ is the Dowling geometry of rank n [3], [19, Section 1]. Deleting the joint h_i from $\mathsf{Q}_n(G;X)$ gives $\mathsf{Q}_n(G;X \setminus \{i\})$.

For $m = 1$, the matroid $\mathsf{Q}_n(1;X)$ is the cycle matroid of the graph on $\{0, 1, \dots, n\}$ with all edges ij, $1 \le i < j \le n$, and the edges $0i$, $i \in X$: identifying h_i with $0i$, the balanced components of $([n], S)$ are exactly the components of the corresponding subgraph that do not contain 0, and (3.1) becomes the graphic rank function. In particular $\mathsf{Q}_n(1) \cong \mathsf{K}_{n+1}$.

3.2. **Potentials.**

Lemma 3.1. *Let $A \subseteq [n]$ be nonempty.*

(a) *Let S be a set of links with both ends in A such that (A, S) is connected and balanced. Then there is $\varphi\colon A \to G$ with $g = \varphi(i)^{-1}\varphi(j)$ for every $e_{ij}(g) \in S$, and φ is unique up to replacing φ by $c\varphi$, $c \in G$.*

(b) *For $\varphi\colon A \to G$ the set $S_\varphi(A) = \{e_{ij}(\varphi(i)^{-1}\varphi(j)) : i < j,\ i, j \in A\}$ is balanced, and $S_\varphi(A) = S_{\varphi'}(A)$ if and only if $\varphi' = c\varphi$ for some $c \in G$. There are exactly $m^{|A|-1}$ distinct sets $S_\varphi(A)$.*

Proof. (a) Fix $i_0 \in A$. For $i \in A$ let $\varphi(i)$ be the product of the gains along a walk from i_0 to i in (A, S). In a balanced gain graph every closed walk has gain 1 (cancel backtracks and split the walk into circles), so φ is well defined, and for a link from i to j with gain g we get $\varphi(j) = \varphi(i)g$. If φ' is another such map, then $\varphi'(i)\varphi(i)^{-1} = \varphi'(j)\varphi(j)^{-1}$ for every link $e_{ij}(g) \in S$, so $\varphi'\varphi^{-1}$ is constant on the connected graph (A, S).

(b) The gain of a circle in $S_\varphi(A)$ is a telescoping product equal to 1. If $S_\varphi(A) = S_{\varphi'}(A)$, then $\varphi'(i)\varphi(i)^{-1} = \varphi'(j)\varphi(j)^{-1}$ for all $i, j \in A$. Finally, G

acts freely on the set G^A of maps by left multiplication, so there are $m^{|A|}/m$ orbits. No commutativity is needed. □

3.3. **Flats.** For $B \subseteq [n]$ let $E(B)$ be the set of all elements of E with all ends in B, that is, the links $e_{ij}(g)$ with $i, j \in B$ and the joints h_i with $i \in X \cap B$. We call B *admissible* if $B = \emptyset$ or the gain graph $(B, E(B))$ is unbalanced. Since $(B, E(B))$ is connected, $B \neq \emptyset$ is admissible if and only if $B \cap X \neq \emptyset$, or $m \geq 2$ and $|B| \geq 2$; in the latter case two links $e_{ij}(g), e_{ij}(g')$ with $g \neq g'$ form an unbalanced circle.

Lemma 3.2. *A set $F \subseteq E$ is a flat of $\mathsf{Q}_n(G; X)$ if and only if*

$$F = E(B_0) \cup S_{\varphi_1}(B_1) \cup \cdots \cup S_{\varphi_b}(B_b) \tag{3.2}$$

for an admissible set B_0, a partition $\{B_1, \dots, B_b\}$ of $[n] \setminus B_0$ into nonempty blocks, and maps $\varphi_l \colon B_l \to G$. The set B_0, the blocks B_l and the classes of the φ_l modulo left multiplication are determined by F, and $\operatorname{rk} F = n - b$.

We call B_0 the *zero block* and $B_1, \dots, B_b$ the *balanced blocks* of F.

Proof. Let F be a flat. If $([n], F)$ had two unbalanced components, a link joining them would not change $b(F)$, hence would lie in the closure of F; this is impossible. Let B_0 be the vertex set of the unbalanced component if it exists and $B_0 = \emptyset$ otherwise. Adding to F an element of $E(B_0)$ does not change the components or their balance, so $E(B_0) \subseteq F$, and B_0 is admissible. Let $B_1, \dots, B_b$ be the vertex sets of the balanced components. By Lemma 3.1(a) there are φ_l with $F \cap E(B_l) \subseteq S_{\varphi_l}(B_l)$; adding an element of $S_{\varphi_l}(B_l)$ keeps B_l balanced, so equality holds. Since every element of F lies inside a component, F has the form (3.2), and $\operatorname{rk} F = n - b$.

Conversely, let F be given by (3.2). The components of $([n], F)$ are B_0 (if nonempty), which is unbalanced, and $B_1, \dots, B_b$, which are balanced; so $\operatorname{rk} F = n - b$. Let $e \in E \setminus F$. If e is a link between two different blocks, then $F \cup \{e\}$ has one balanced component fewer: either a balanced block merges with B_0, or two balanced blocks merge into a balanced component, since e is a bridge. If e is a link inside B_l not in $S_{\varphi_l}(B_l)$, it forms an unbalanced circle of length two with the link of $S_{\varphi_l}(B_l)$ on the same pair. If e is a joint at a vertex of B_l, the component B_l becomes unbalanced. In each case $\operatorname{rk}(F \cup \{e\}) > \operatorname{rk} F$, so F is a flat. The uniqueness statement follows from Lemma 3.1. □

Corollary 3.3. *Let $n \geq 1$. The rank of $\mathsf{Q}_n(G; X)$ is n, except when $X = \emptyset$ and either $m = 1$ or $n = 1$, in which case it is $n - 1$. If $m = 1$, then $\mathsf{Q}_n(1; \emptyset) \cong \mathsf{K}_n$. For $n = 0$ the matroid is empty and has rank 0.*

Proof. The set E is the flat with $B_0 = [n]$ if $[n]$ is admissible, that is, if $X \neq \emptyset$ or $m \geq 2$ and $n \geq 2$. Otherwise $E = S_1([n])$ is a single balanced block, of rank $n - 1$; for $m = 1$ the matroid is then the cycle matroid of K_n. □

3.4. **Restrictions and contractions.**

Lemma 3.4. *Let F be a flat as in* (3.2)*. Then*

$$\mathsf{Q}_n(G;X)|F \;\cong\; \mathsf{Q}_{|B_0|}(G;X\cap B_0)\oplus \mathsf{K}_{|B_1|}\oplus\cdots\oplus \mathsf{K}_{|B_b|},$$

where the first summand is $\mathsf{Q}_n(G;X)|E(B_0)$, read on the vertex set B_0, and K_1 is the empty matroid.

Proof. No element of F joins two different blocks, so for $S\subseteq F$ the balanced components of $([n],S)$ are obtained separately inside each block; hence rk is additive over the blocks. The restriction to $E(B_0)$ is the frame matroid on B_0 described in the statement. On $S_{\varphi_l}(B_l)$ every subset is balanced, so the rank function is that of the cycle matroid of the complete graph on B_l. □

Lemma 3.5. *Let F be a flat as in* (3.2)*, and suppose that $B_0\neq\emptyset$ or $X=[n]$. Then $\mathsf{Q}_n(G;X)/F$ is loopless and* $\mathrm{si}(\mathsf{Q}_n(G;X)/F)\cong\mathsf{Q}_b(G)$*. In particular, if $i\in X$, then $\{h_i\}$ is a flat and* $\mathrm{si}(\mathsf{Q}_n(G;X)/h_i)\cong\mathsf{Q}_{n-1}(G)$*.*

Proof. Let $M=\mathsf{Q}_n(G;X)$ and $\varphi\colon [n]\setminus B_0\to G$ the map that restricts to φ_l on B_l. Let $M'=\mathsf{Q}_b(G)$ on the vertex set $[b]$. Define $\psi\colon E\setminus F\to E(M')$ as follows. A link $e_{ij}(g)$ with $i\in B_p$, $j\in B_q$, $p\neq q$, is sent to the link between p and q that has gain $\varphi(i)g\varphi(j)^{-1}$ when traversed from p to q. A link inside some B_p (necessarily not in $S_{\varphi_p}(B_p)$), a link between B_0 and B_p, and a joint at a vertex of B_p are sent to the joint h_p. Elements of $E(B_0)$ lie in F. Every link of M' lies in the image, since g can be chosen freely; every joint h_p lies in the image because $B_0\neq\emptyset$ or B_p contains a joint.

We claim that $\mathrm{rk}_M(S\cup F)-\mathrm{rk}_M(F)=\mathrm{rk}_{M'}(\psi(S))$ for all $S\subseteq E\setminus F$. Replacing every gain g of a link from i to j by $\varphi(i)g\varphi(j)^{-1}$, for $i,j\notin B_0$, does not change the balance of any circle (circle gains are conjugated), and makes all links of F outside B_0 have gain 1. Every component of $([n],S\cup F)$ is a union of blocks. The component containing B_0, if any, is unbalanced; its blocks other than B_0 form components of $([b],\psi(S))$ each of which contains a joint coming from a link to B_0, hence is unbalanced. A component C not meeting B_0, formed by blocks B_p, $p\in P$, is balanced if and only if it has a potential, which must be constant on every B_p since the links of F have gain 1; this happens if and only if S contains no joint and no link inside a block, and the links between blocks admit a potential on P, that is, if and only if the component of $([b],\psi(S))$ on P is balanced. Hence $b(S\cup F)=b_{M'}(\psi(S))$, and the claim follows from (3.1) since $b(F)=b$.

By the claim, every element of M/F has rank one, and two elements are parallel if and only if they have the same image under ψ, because M' is simple. Since ψ is surjective, $\mathrm{si}(M/F)\cong M'$. The last statement is the case $B_0=\{i\}$, with all other blocks singletons. □

3.5. **Deleting a joint.** For a finite group G of order m and $j\geq 0$ we write

$$Y_j(t)=Y_{\mathsf{Q}_j(G)}(t),\qquad Y_0=1,$$

and for $h \geq 1$

$$\tau_h = \tau\big(\mathsf{Q}_{2h-1}(G)\big) = m^{2h-2}\,(2h-3)!!\,(2h-1)^{h-2}, \tag{3.3}$$

where $(-1)!! = 1$. The formula (3.3) is [6, Example 4.2]; for $h = 1$ it is the fact that $P = 1$ for a matroid of rank one. For $m = 1$ it gives the leading coefficient of $P_{\mathsf{K}_{2h}}$, conjectured in [4] and proved in [5]; see also [9, Theorem 2].

Proposition 3.6. *Let $n \geq 2$, $X \subseteq [n]$, $k = |X|$, $i \in X$, $M = \mathsf{Q}_n(G; X)$ and $e = h_i$. Assume that $m \geq 2$ or $k \geq 2$. Then e is not a coloop of M, and*

$$Y_M(t) - Y_{M\setminus e}(t) = (1+t)\,Y_{n-1}(t) - \sum_{h=1}^{\lfloor n/2 \rfloor} N_h\,\tau_h\,t^h\,Y_{n-2h}(t),$$

where $N_h = \binom{n-1}{2h-1}$ if $m \geq 2$, and $N_h = \binom{n-1}{2h-1} - \binom{n-k}{2h-1}$ if $m = 1$.

Proof. If $m \geq 2$, choose $j \neq i$ and two links between i and j with different gains; they span a set of rank 2 which contains e in its closure. If $m = 1$, choose $j \in X \setminus \{i\}$; then $\{h_j, e_{ij}(1)\}$ has rank 2 and spans e. In both cases e is not a coloop.

By Lemma 3.5, $\mathrm{si}(M/e) \cong \mathsf{Q}_{n-1}(G)$, so $Y_{M/e} = Y_{n-1}$ by Lemma 2.2(a). Let F be a flat with zero block B_0 and balanced blocks $B_1, \dots, B_b$. Then $e \in F$ if and only if $i \in B_0$. In that case $F \setminus \{e\}$ is the union of $E(B_0) \setminus \{e\}$ and the sets $S_{\varphi_l}(B_l)$. If the gain graph $(B_0, E(B_0)\setminus\{e\})$ is unbalanced, then adding e to $F \setminus \{e\}$ does not change b, so $F \setminus \{e\}$ is not a flat. Otherwise $E(B_0) \setminus \{e\} = S_1(B_0)$ (all gains are trivial and there are no joints, or $B_0 = \{i\}$), and $F \setminus \{e\}$ is a flat by Lemma 3.2, with B_0 turned into a balanced block. Hence $F \in \mathcal{T}_e$ if and only if $i \in B_0$ and

$$|B_0| \geq 2 \quad \text{if } m \geq 2, \qquad (B_0 \cap X) \setminus \{i\} \neq \emptyset \quad \text{if } m = 1.$$

Let $F \in \mathcal{T}_e$ and $z = |B_0| \geq 2$. By Lemma 3.4, $M|F/e$ is the direct sum of $\mathsf{Q}_z(G; X \cap B_0)/h_i$ and the matroids $\mathsf{K}_{|B_l|}$, and by Lemma 3.5 the first summand simplifies to $\mathsf{Q}_{z-1}(G)$, which has positive rank. By Lemma 2.2(c), $\tau(M|F/e) = 0$ unless all balanced blocks are singletons. In that case $M|F/e$ simplifies to $\mathsf{Q}_{z-1}(G)$, so $\tau(M|F/e) \neq 0$ forces $z = 2h$ even, and then $\tau(M|F/e) = \tau_h$, $\operatorname{rk} F = n - b = 2h$, and $\mathrm{si}(M/F) \cong \mathsf{Q}_{n-2h}(G)$ by Lemma 3.5. A flat with singleton balanced blocks is determined by B_0, since a map on a single vertex is unique up to left multiplication. The number of sets $B_0 \ni i$ of size $2h$ satisfying the condition above is $\binom{n-1}{2h-1}$ for $m \geq 2$, and $\binom{n-1}{2h-1} - \binom{n-k}{2h-1}$ for $m = 1$, where the subtracted term counts the sets B_0 with $B_0 \setminus \{i\} \subseteq [n] \setminus X$. Theorem 2.3 now gives the formula. □

4. Dowling geometries

Throughout this section G is a finite group of order m, $a = 1/m$, and $Y_n = Y_{\mathsf{Q}_n(G)}$.

4.1. **The convolution identity.** Put

$$\mathcal{A}_G(x) = \sum_{n\geq 0} Z_{\mathsf{Q}_n(G)}(t)\frac{x^n}{n!}, \qquad \mathcal{D}(x) = \sum_{n\geq 0}(-1)^n Y_n(t)\frac{x^n}{n!},$$
$$\Phi(x) = \sum_{u\geq 1} m^{u-1} Z_{\mathsf{K}_u}(t)\frac{x^u}{u!}.$$

Proposition 4.1. $\mathcal{A}_G(x)\,\mathcal{D}(\Phi(x)) = 1$.

Proof. Let $n \geq 1$ and $M = \mathsf{Q}_n(G)$. Since $X = [n]$, every subset of $[n]$ is admissible. By Lemmas 3.2, 3.1(b), 3.4, 3.5 and 2.2, the flats F of M with zero block B_0 and balanced blocks $B_1, \ldots, B_b$ are $\prod_l m^{|B_l|-1}$ in number, and each of them has

$$Z_{M|F} = Z_{\mathsf{Q}_{|B_0|}(G)} \prod_{l=1}^{b} Z_{\mathsf{K}_{|B_l|}}, \qquad Y_{M/F} = Y_b, \qquad \operatorname{rk} M - \operatorname{rk} F = b.$$

Thus (2.2) reads

$$\sum_{B_0\subseteq[n]} Z_{\mathsf{Q}_{|B_0|}(G)} \sum_{b\geq 0}(-1)^b Y_b \sum_{\{B_1,\ldots,B_b\}} \prod_{l=1}^{b} m^{|B_l|-1} Z_{\mathsf{K}_{|B_l|}} = 0,$$

where the inner sum is over the partitions of $[n] \setminus B_0$ into b blocks. For $n = 0$ the left side equals 1. By the exponential formula [17, Section 5.1], the exponential generating function of the inner sum is $\Phi(x)^b/b!$, and the sum over B_0 is a binomial convolution. This proves the proposition. □

4.2. **The generating function of Ferroni and Larson.** Let

$$g_t(u) = \frac{\log(1+tu)}{t} + \log(1+u) - u = \sum_{i\geq 1} \frac{(-1)^{i-1}(t^{i-1}+1)}{i} u^i - u \in \mathbb{Q}[t][[u]].$$

Since $g_t(u) = u + O(u^2)$, it has a compositional inverse $g_t^{\langle -1\rangle}$ in $\mathbb{Q}[t][[u]]$. Put

$$\mathcal{C}(x) = (1+t)x + t\int_0^x g_t^{\langle -1\rangle}(s)\,ds, \qquad \mathcal{A}(x) = e^{\mathcal{C}(x)}, \qquad \mathcal{H}(x) = \int_0^x \mathcal{A}(s)\,ds.$$

Proposition 4.2. (a) $n!\,[x^n]\,\mathcal{A}(x) = Z_{\mathsf{K}_{n+1}}(t)$ *for all* $n \geq 0$*; consequently* $\mathcal{H}(x) = \sum_{u\geq 1} Z_{\mathsf{K}_u}(t)x^u/u!$ *and* $\Phi(x) = \mathcal{H}(mx)/m$.

(b) $\mathcal{A}_G(x) = \mathcal{A}(mx)^{1/m}$.

Proof. (a) By [5, Propositions 2.3 and 2.8], $\mathcal{A}(x) = \sum_{n,k} \alpha(n,k)\frac{x^n}{n!}t^k$, where $\alpha(n,k)$ is the number of quasi series-parallel matroids on $[n]$ of rank k, and [5, Corollary 3.2] states $n![x^n]\mathcal{A}(x) = Z_{\mathsf{K}_{n+1}}(t)$. The remaining statements follow by integration.

(b) By [6, Theorem 1.1] and the enumerative form of it stated after that theorem, the coefficient of t^i in $Z_{\mathsf{Q}_n(G)}(t)$ is the number $\alpha_G(n, n-i)$ of

equivalence classes of G-labeled quasi series-parallel matroids on $[n]$ of rank $n-i$, and by [6, Proposition 3.2], $\sum_{n,k} \alpha_G(n,k)\frac{x^n}{n!}t^k = \mathcal{A}(mx)^{1/m}$. Hence

$$\mathcal{A}(mx)^{1/m} = \sum_{n\geq 0} \frac{x^n}{n!}\, t^n\, Z_{\mathsf{Q}_n(G)}(t^{-1}) = \mathcal{A}_G(x),$$

because $Z_{\mathsf{Q}_n(G)}$ is palindromic of degree n. □

Lemma 4.3. *$\mathcal{A}(g_t(u)) = (1+u)(1+tu)\, e^{-tu^2/2}$ and $\mathcal{H}(g_t(u)) = u\, e^{-tu^2/2}$.*

Proof. We have $g_t'(u) = \frac{1}{1+tu} + \frac{1}{1+u} - 1 = \frac{1-tu^2}{(1+u)(1+tu)}$ and $\mathcal{C}'(x) = 1 + t + t\, g_t^{\langle -1\rangle}(x)$, so

$$\frac{d}{du}\mathcal{C}(g_t(u)) = \frac{(1+t+tu)(1-tu^2)}{(1+u)(1+tu)} = \frac{1}{1+u} + \frac{t}{1+tu} - tu,$$

where the second equality follows from $(1+t+tu)(1-tu^2) = (1+tu)+t(1+u)-tu(1+u)(1+tu)$. Both $\mathcal{C}(g_t(u))$ and $\log(1+u)+\log(1+tu)-tu^2/2$ vanish at $u=0$, so they are equal, which gives the first identity. Then

$$\frac{d}{du}\mathcal{H}(g_t(u)) = \mathcal{A}(g_t(u))\, g_t'(u) = (1-tu^2)e^{-tu^2/2} = \frac{d}{du}\big(ue^{-tu^2/2}\big),$$

and both sides of the second identity vanish at $u=0$. □

4.3. **Proof of Theorem 1.4.** By Propositions 4.1 and 4.2, $\mathcal{D}(\mathcal{H}(mx)/m) = \mathcal{A}(mx)^{-a}$. Substituting $x = g_t(u)/m$ and using Lemma 4.3,

$$\mathcal{D}\Big(\frac{u\, e^{-tu^2/2}}{m}\Big) = (1+u)^{-a}(1+tu)^{-a}e^{atu^2/2}.$$

Now substitute $u = -v$, where $ve^{-tv^2/2} = mx$. The argument of $\mathcal{D}$ becomes $-x$, and $\mathcal{D}(-x) = \sum_n Y_n x^n/n!$. This proves the first statement of Theorem 1.4.

For the coefficients, write $z = mx$, so that $v = ze^{tv^2/2}$. By (2.3) with $F(u) = e^{atu^2/2}(1-u)^{-a}(1-tu)^{-a}$,

$$\frac{Y_n(t)}{n!\, m^n} = [u^n]\,(1-tu^2)\, e^{\lambda tu^2}(1-u)^{-a}(1-tu)^{-a}, \qquad \lambda = \frac{n+a}{2}. \tag{4.1}$$

By (2.5), the coefficient of t^p on the right is $[\xi^{n-p}\eta^p]\,(1-z)e^{\lambda z}R^{-a}$. Expanding with (2.6) and (2.4) gives the second statement of Theorem 1.4. □

Example 4.4. For small n, Theorem 1.4 gives

$$\begin{aligned}
Y_{\mathsf{Q}_1(G)} &= 1+t,\\
Y_{\mathsf{Q}_2(G)} &= (m+1) + (m+2)t + (m+1)t^2,\\
Y_{\mathsf{Q}_3(G)} &= (m+1)(2m+1)(1+t^3) + 3(m+1)^2(t+t^2),\\
Y_{\mathsf{Q}_4(G)} &= (m+1)(2m+1)(3m+1)(1+t^4)\\
&\quad + 2(m+1)(2m+1)(3m+2)(t+t^3)\\
&\quad + 3(4m^3+11m^2+8m+2)\, t^2.
\end{aligned}$$

Since $\mathsf{Q}_2(G)$ is the uniform matroid $\mathsf{U}_{2,m+2}$, the case $n = 2$ agrees with [10, Theorem 1.1].

4.4. **Raised weights and log-concavity.**

Proposition 4.5. *Let $n \geq 0$, $\lambda = (n+a)/2$, $J = \lfloor n/2 \rfloor$, $d_l = c_l(a+1)$, and*

$$U_j = \frac{a\,(\lambda - j)\,\lambda^{j-2}}{2 \cdot j!} \qquad (0 \leq j \leq J). \tag{4.2}$$

Then $[t^p]\,Y_n = n!\,m^n \sum_{j=0}^{\min(p,n-p)} U_j\, d_{p-j}\, d_{n-p-j}$ for $0 \leq p \leq n$. Consequently Y_n has degree n, positive palindromic coefficients, and $Y_n(0) = \prod_{i=0}^{n-1}(1+im)$.

Proof. Apply Corollary 2.5 to $V(z) = (1-z)e^{\lambda z}$ and $\beta = a$; here $n+a-2j = 2(\lambda - j) > 0$ for $j \leq J$. Using $w_j(\lambda)/(\lambda - j) = \lambda^{j-1}/j!$ for $j \geq 0$, the weights are

$$\frac{a}{2}\Big(\frac{\lambda^{j-1}}{j!} - \frac{\lambda^{j-2}}{(j-1)!}\Big) = U_j \quad (j \geq 1), \qquad \frac{a}{2\lambda} = U_0.$$

The term $j = 0$ is positive for every p, and the expression is symmetric under $p \mapsto n - p$. Finally $Y_n(0) = n!m^n U_0 d_n = m^n a(a+1)\cdots(a+n-1) = \prod_{i=0}^{n-1}(1+im)$. □

Lemma 4.6. *Let $n \geq 2$, $1 \leq j \leq \lfloor n/2 \rfloor - 1$ and $r = \lambda - j$. Then $r \geq 1 + a/2$ and*

$$\log \frac{U_j^2}{U_{j-1}U_{j+1}} = \log\Big(1 + \frac{1}{j}\Big) - \log\Big(1 - \frac{1}{r^2}\Big) \geq \frac{1}{j+1} + \frac{1}{r^2}.$$

Proof. Since $j \leq \lfloor n/2 \rfloor - 1$, we have $r \geq \frac{n+a}{2} - \lfloor \frac{n}{2} \rfloor + 1 \geq 1 + \frac{a}{2}$. The indices $j-1, j, j+1$ correspond to $r+1, r, r-1$, and a direct cancellation in (4.2) gives $U_j^2/(U_{j-1}U_{j+1}) = \frac{j+1}{j} \cdot \frac{r^2}{r^2-1}$. The inequality follows from $\log(1+x) \geq x/(1+x)$ and $-\log(1-x) \geq x$. □

Theorem 4.7. *For every finite group G and every $n \geq 2$, the coefficients of $Y_{\mathsf{Q}_n(G)}$ are positive and strictly log-concave. In particular this holds for the braid matroids K_{n+1} and for the matroids of type B_n.*

Proof. By Lemma 4.6 the weights U_j are positive and log-concave, so Lemma 2.8 with $\beta = a$ applies to Proposition 4.5. □

5. Nontrivial groups

In this section G is a finite group of order $m \geq 2$, so that $0 < a = 1/m \leq \frac{1}{2}$. We keep the notation $Y_j = Y_{\mathsf{Q}_j(G)}$ and τ_h from (3.3).

5.1. **Interpolation.**

Proposition 5.1. *Let $n \geq 2$, $X \subseteq [n]$ and $k = |X|$. Put*

$$\Delta_n(t) = (1+t)\,Y_{n-1}(t) - \sum_{h=1}^{\lfloor n/2 \rfloor} \binom{n-1}{2h-1} \tau_h\, t^h Y_{n-2h}(t), \qquad Y_n^\circ = Y_n - n\Delta_n.$$

Then $Y_{\mathsf{Q}_n(G;X)} = Y_n - (n-k)\Delta_n = \frac{k}{n}\,Y_n + \big(1 - \frac{k}{n}\big)Y_n^\circ$. In particular $Y_n^\circ = Y_{\mathsf{Q}_n(G;\emptyset)}$.

Proof. Since $m \geq 2$, Proposition 3.6 applies to every joint, and the increment it gives does not depend on X. Deleting the joints of $\mathsf{Q}_n(G)$ one at a time down to X gives the first formula, and the second is a rearrangement. □

5.2. **The jointless series.**

Lemma 5.2. *Let v be as in Theorem* 1.4. *Then*

$$T(x) := \sum_{h \geq 1} \tau_h\, t^h \frac{x^{2h-1}}{(2h-1)!} = \frac{t}{m}\Big(v - \frac{tv^3}{3}\Big).$$

Proof. Write $z = mx$ and $h = l+1$. Since $(2l+1)! = 2^l\, l!\,(2l+1)!!$, (3.3) gives $\tau_{l+1}/(2l+1)! = m^{2l}(2l+1)^{l-2}/(2^l\, l!)$. On the other hand, by (2.3), $[z^{2l+1}](v - tv^3/3)$ is t^l times the coefficient of y^l in $(1 - \frac{y}{3})(1-y)e^{\gamma y}$ with $\gamma = l + \frac{1}{2}$, namely

$$\frac{\gamma^{l-2}}{l!}\Big(\gamma^2 - \frac{4}{3}\,l\gamma + \frac{1}{3}\,l(l-1)\Big) = \frac{\gamma^{l-2}}{4 \cdot l!} = \frac{(2l+1)^{l-2}}{2^l\, l!}.$$

The series v is odd in x, so the even coefficients of both sides vanish, and the odd coefficients agree after multiplying by t/m and m^{2l+1}. □

Proposition 5.3. *Let $\mathcal{F}(x) = \sum_{n \geq 0} Y_n\, x^n/n!$. For every $n \geq 2$,*

$$\frac{Y_n^\circ}{n!} = [x^n]\,\mathcal{F}(x)\Big(1 - (1+t)x + \frac{tx}{m}\Big(v - \frac{tv^3}{3}\Big)\Big).$$

Proof. We have $n(1+t)Y_{n-1}/n! = [x^n]\,(1+t)x\mathcal{F}(x)$ and $n\binom{n-1}{2h-1}/n! = 1/((2h-1)!\,(n-2h)!)$, so $n\Delta_n/n! = [x^n]\,(1+t)x\mathcal{F}(x) - [x^n]\,xT(x)\mathcal{F}(x)$. Lemma 5.2 completes the proof. □

The series in Proposition 5.3 has constant coefficient 1, which agrees with $Y_{\mathsf{Q}_0(G;\emptyset)} = 1$, but its linear coefficient is 0, whereas $Y_{\mathsf{Q}_1(G;\emptyset)} = 1$, because the joint of $\mathsf{Q}_1(G)$ is a coloop. We use the series only for $n \geq 2$.

5.3. **The raised weights.** For the rest of this section fix $n \geq 2$ and put

$$\lambda = \frac{n+a}{2}, \quad \mu = \lambda - \frac{1}{2} = \frac{n+a-1}{2}, \quad s = \frac{\mu}{\lambda}, \quad J = \Big\lfloor \frac{n}{2} \Big\rfloor, \quad r_j = \lambda - j,$$

and let U_j and $d_l = c_l(a+1)$ be as in Proposition 4.5. Let

$$\gamma_0 = \frac{an}{n+a-1}, \qquad \gamma_1 = a(1-a), \qquad \gamma_2 = \frac{a^2}{3},$$

$$V(z) = (1-z)\Big(e^{\lambda z} - (\gamma_0 + \gamma_1 z + \gamma_2 z^2)e^{\mu z}\Big), \qquad v_j = [z^j]\,V(z),$$

and $v_{-1} = 0$.

Proposition 5.4. *For $0 \le j \le J$ put $W_j = \frac{a}{2}\big(\frac{v_j}{\lambda - j} - \frac{v_{j-1}}{\lambda - j + 1}\big)$. Then*

$$[t^p]\,Y_n^\circ = n!\,m^n \sum_{j=0}^{\min(p,n-p)} W_j\, d_{p-j}\, d_{n-p-j} \qquad (0 \le p \le n).$$

Proof. Put $z = mx$, so that $v = ze^{tv^2/2}$. Writing u for v, we have $x = a\,u\,e^{-tu^2/2}$, and the bracket in Proposition 5.3 becomes

$$1 - e^{-tu^2/2}\Big(a(1+t)u - a^2tu^2 + \frac{a^2t^2u^4}{3}\Big).$$

By (2.3), as in (4.1), $Y_n^\circ/(n!m^n)$ is the coefficient of u^n in

$$(1-tu^2)(1-u)^{-a}(1-tu)^{-a}\Big(e^{\lambda tu^2} - e^{\mu tu^2}\Big(a(1+t)u - a^2tu^2 + \frac{a^2t^2u^4}{3}\Big)\Big).$$

With $\xi = u$, $\eta = tu$ and (2.5), the coefficient of t^p is $[\xi^{n-p}\eta^p]$ of

$$K_a(\lambda) - \Big(a(\xi+\eta) - a^2z + \frac{a^2}{3}z^2\Big)K_a(\mu), \qquad K_\beta(\theta) = (1-z)e^{\theta z}R^{-\beta}.$$

Write $\xi + \eta = 1 + z - R$, so that $(\xi+\eta)K_a(\mu) = (1+z)K_a(\mu) - K_{a-1}(\mu)$. Since $2\mu = n + (a-1)$ and $n + a - 1 > 0$, Lemma 2.4(b) gives $[K_{a-1}(\mu)]_n = \frac{a-1}{n+a-1}[K_a(\mu)]_n$. Hence the component of degree n equals that of

$$K_a(\lambda) - \Big(a + \frac{a(1-a)}{n+a-1} + a(1-a)z + \frac{a^2}{3}z^2\Big)K_a(\mu) = V(z)\,R^{-a},$$

since $a + \frac{a(1-a)}{n+a-1} = \gamma_0$. Corollary 2.5 with $\beta = a$ completes the proof; note $n + a - 2j = 2(\lambda - j) > 0$. □

We next compute W_j in closed form. For $0 < r$ put

$$C(r) = \frac{a^2r^3}{3} + \Big(a^2 - \Big(a - \frac{a^2}{3}\Big)\mu\Big)r^2 + \Big(\Big(2a - \frac{2a^2}{3}\Big)\mu^2 - 2a^2\mu + \frac{a^2}{6}\Big)r$$
$$+ \frac{a(1-a)\mu}{4} - \frac{a^2}{2} + \frac{a^2(3r-2\mu)}{16\,r(r+1)}, \tag{5.1}$$

and $\Psi(r) = C(r)/r$.

Proposition 5.5. *For $0 \le j \le J$ we have $W_j = U_j q_j$ with*

$$1 - q_j = \frac{s^{j-2}\,\Psi(r_j)}{\mu^2}. \tag{5.2}$$

Proof. Fix j and write $r = r_j = \lambda - j$, so that $j = \mu + \frac{1}{2} - r$. For $i \ge 0$ we have $w_i(\theta) = (\theta - i)\theta^{i-1}/i!$, and $w_{-1} = w_{-2} = 0$. Hence, for every $j \ge 0$,

$$\gamma_0 w_j(\mu) + \gamma_1 w_{j-1}(\mu) + \gamma_2 w_{j-2}(\mu) = \frac{\mu^{j-3}}{j!}\,N(r),$$

where

$$N(r) = \gamma_0\mu^2\Big(r-\frac{1}{2}\Big) + \gamma_1\mu\, j\Big(r+\frac{1}{2}\Big) + \gamma_2\, j(j-1)\Big(r+\frac{3}{2}\Big), \qquad j = \mu+\tfrac{1}{2}-r;$$

the factors j and $j(j-1)$ vanish exactly when the corresponding w has negative index. Thus $v_j = w_j(\lambda) - \mu^{j-3}N(r)/j!$, and $v_{j-1} = w_{j-1}(\lambda) - \mu^{j-4}N(r+1)/(j-1)!$ for $j \geq 1$, where $N(r+1)$ is computed with $j-1 = \mu+\frac{1}{2}-(r+1)$. The contribution of $w(\lambda)$ to W_j is $U_j = \frac{a}{2}\cdot\frac{r\lambda^{j-2}}{j!}$, as in the proof of Proposition 4.5. Therefore

$$W_j = U_j - \frac{a\,\mu^{j-4}}{2\cdot j!}\Big(\frac{\mu\,N(r)}{r} - \frac{j\,N(r+1)}{r+1}\Big),$$

which is valid also for $j = 0$. Using $\gamma_0\mu^2 = \frac{1}{2}a\mu(2\mu+1-a)$, a direct expansion shows that the bracket equals $C(r)$. Dividing by U_j gives (5.2). □

We record two consequences of (5.1) that are checked by direct expansion. First,

$$\Psi(r) = p(r) + \frac{\nu}{r} + \frac{3a^2}{16}\cdot\frac{1}{r(r+1)} - \frac{a^2\mu}{8}\,\varrho(r), \qquad \varrho(r) = \frac{1}{r^2(r+1)}, \tag{5.3}$$

where $\nu = \frac{a(1-a)\mu}{4} - \frac{a^2}{2}$ and

$$p(r) = \frac{a^2r^2}{3} + \Big(a^2 - \Big(a-\frac{a^2}{3}\Big)\mu\Big)r + \Big(2a - \frac{2a^2}{3}\Big)\mu^2 - 2a^2\mu + \frac{a^2}{6}.$$

Second, Ψ is a polynomial of degree two in a for fixed μ and r: writing Ψ_a to indicate the dependence on a, we have $\Psi_a = a\Psi^{(1)} + a^2\Psi^{(2)}$ with

$$\Psi^{(1)} = \mu\Big(2\mu - r + \frac{1}{4r}\Big), \tag{5.4}$$

$$\Psi^{(1)} + \Psi^{(2)} = \mu^2 + \frac{(\mu-r)^2}{3} + r - 2\mu + \frac{1}{6} - \frac{1}{2r} + \frac{3}{16r(r+1)} - \frac{\mu}{8r^2(r+1)}.$$

Here μ and r are treated as independent variables.

5.4. **Positivity of the weights.**

Lemma 5.6. *Let q_j be as in Proposition 5.5.*

(a) $q_0 = \dfrac{(1-a)(n-1)}{n+a-1} > 0.$

(b) *With $\varsigma = \mu - \frac{1}{2} = \frac{n+a-2}{2} > 0$,*

$$q_1 = 1 - a - \frac{a(1-\frac{3}{2}a)}{\varsigma} + \frac{a^2}{2\varsigma^2} > 0,$$

and $q_1 > 1 - 2a + \frac{3}{2}a^2$ if $n \geq 4$.

(c) *$q_j > \frac{1}{6}$ for $2 \leq j \leq J$.*

In particular $W_j > 0$ for $0 \leq j \leq J$.

Proof. (a) and the formula in (b) follow from (5.2) and (5.1) by substitution, or directly from the definition of W_0 and W_1. As a quadratic polynomial in $1/\varsigma$, q_1 has positive leading coefficient and discriminant $a^2(\frac{9}{4}a^2 - a - 1) < 0$ for $0 < a \le \frac{1}{2}$; hence $q_1 > 0$. If $n \ge 4$, then $\varsigma \ge 1 + \frac{a}{2} > 1$; since $1 - \frac{3}{2}a > 0$, dropping the last term gives $q_1 > 1 - a - a(1 - \frac{3}{2}a)$.

(c) Let $2 \le j \le J$ and $r = r_j$. Then $\mu = r + j - \frac{1}{2} \ge r + \frac{3}{2}$. A direct expansion of (5.1) gives, for every $r > 0$,

$$\Big(2a - \frac{2a^2}{3}\Big)\mu^2 - \Psi_a(r) = a^2\Big(4\Big(\frac{5}{6}\mu^2 - \Psi_{1/2}(r)\Big) + \Big(\frac{1}{a} - 2\Big)\mu\Big(r - \frac{1}{4r}\Big)\Big)$$

and

$$\frac{5}{6}\mu^2 - \Psi_{1/2}(r) = \frac{r(5\mu - 3 - r)}{12} + \Big(\frac{\mu}{2} - \frac{1}{24} - \frac{\mu}{16r}\Big) + \Big(\frac{1}{8r} - \frac{3r - 2\mu}{64r^2(r+1)}\Big).$$

Suppose first $r \ge \frac{1}{2}$. The three groups on the right of the second identity are positive: $5\mu - 3 - r \ge 4\mu - \frac{3}{2} > 0$; $\frac{\mu}{2} - \frac{1}{24} - \frac{\mu}{16r} \ge \frac{3\mu}{8} - \frac{1}{24} > 0$; and $\frac{3r-2\mu}{64r^2(r+1)} \le \frac{3}{64r(r+1)} < \frac{1}{8r}$. Since $\frac{1}{a} - 2 \ge 0$ and $r - \frac{1}{4r} \ge 0$, we get $\Psi_a(r) < (2a - \frac{2a^2}{3})\mu^2$. Otherwise $r < \frac{1}{2}$, which happens only for n even and $j = J$, where $r = \frac{a}{2}$. Substituting $r = \frac{a}{2}$ into (5.1),

$$\Psi_a\Big(\frac{a}{2}\Big) = \Big(2a - \frac{2a^2}{3}\Big)\mu^2 + \Big(\frac{a^3}{6} - \frac{5a^2}{2} - \frac{a(1+a)}{2(a+2)}\Big)\mu + a\Big(\frac{a^3}{12} + \frac{a^2}{2} + \frac{a}{6} - 1 + \frac{3}{4(a+2)}\Big).$$

For $0 < a \le \frac{1}{2}$ the coefficient of μ is negative, and the last bracket is at most $\frac{1}{96} + \frac{1}{8} + \frac{1}{12} - 1 + \frac{3}{8} < 0$. Again $\Psi_a(r) < (2a - \frac{2a^2}{3})\mu^2$.

In both cases (5.2), $s^{j-2} \le 1$ and $2a - \frac{2a^2}{3} \le \frac{5}{6}$ give $1 - q_j < \frac{5}{6}$ if $\Psi_a(r) \ge 0$, and $1 - q_j \le 0$ otherwise. □

5.5. **Log-concavity of the weights.** For $1 \le j \le J - 1$ we write $\delta_j = q_{j-1} + q_{j+1} - 2q_j$. For a function f of r let

$$\mathcal{L}_s f(r) = s\, f(r-1) + s^{-1} f(r+1) - 2f(r).$$

Since the indices $j - 1, j, j + 1$ correspond to $r_j + 1, r_j, r_j - 1$, (5.2) gives

$$\delta_j = -\frac{s^{j-2}}{\mu^2}\, \mathcal{L}_s \Psi(r_j). \tag{5.5}$$

Lemma 5.7. *Let $r > 1$.*

(a) $\mathcal{L}_s(1/r) \ge 0$ *and* $\mathcal{L}_s\big(1/(r(r+1))\big) \ge 0$.
(b) *If* $0 < a \le \frac{1}{2}$ *and* $0 < r \le \mu$, *then* $2\mu(2\mu+1)\, \mathcal{L}_s p(r) \ge -a\mu(3\mu+1)$.
(c) $\mathcal{L}_s \varrho(r) \le s\, \varrho(r-1) + s^{-1} \varrho(r+1)$.

Proof. (a) For $r > 0$ we have $\frac{1}{r} = \int_0^1 x^{r-1} dx$ and $\frac{1}{r(r+1)} = \int_0^1 x^{r-1}(1-x)\, dx$. For $r > 1$ these representations hold at $r - 1$ as well, and $\mathcal{L}_s$ multiplies the integrand by $\frac{s}{x} + \frac{x}{s} - 2 = (\sqrt{s/x} - \sqrt{x/s})^2 \ge 0$.

(b) Since $s = \frac{2\mu}{2\mu+1}$, a direct expansion gives

$$2\mu(2\mu+1)\mathcal{L}_s p(r) = \frac{a^2r^2}{3} + \Big((3a^2-a)\mu + \frac{5a^2}{3}\Big)r + \Big(\frac{10a^2}{3} - 2a\Big)\mu^2 + \Big(\frac{11a^2}{3} - a\Big)\mu + \frac{3a^2}{2}.$$

If $3a^2 - a \le 0$, then $(3a^2-a)\mu r \ge (3a^2-a)\mu^2$, and the right side is at least $(\frac{19a^2}{3} - 3a)\mu^2 - a\mu \ge -a\mu(3\mu+1)$. If $3a^2 - a > 0$, the term with r is positive and the right side is at least $-2a\mu^2 - a\mu$.

(c) Drop the term $-2\varrho(r) \le 0$. □

Proposition 5.8. *Let $0 < a \le \frac{1}{2}$, $2 \le j \le J-1$ and $r = r_j$. Then $q_j > \frac{1}{3}$ and $3\delta_j < \frac{1}{j+1} + \frac{1}{r^2}$.*

Proof. Since $J \ge j+1 \ge 3$, we have $n \ge 6$ and

$$\mu \ge \frac{5}{2}, \qquad 1 + \frac{a}{2} \le r \le \mu - \frac{3}{2}, \qquad j+1 \le \frac{n}{2} = \mu + \frac{1-a}{2}. \tag{5.6}$$

The bound $q_j > \frac{1}{3}$. By (5.4), $\Psi^{(1)} > 0$, and using $r \ge 1$ and $\mu - r \ge \frac{3}{2}$,

$$\Psi^{(1)} + \Psi^{(2)} \ge \mu^2 + \frac{3}{4} + 1 - 2\mu + \frac{1}{6} - \frac{1}{2} - \frac{\mu}{16} = \mu^2 - \frac{33}{16}\mu + \frac{17}{12} > 0.$$

Hence $\partial\Psi_\alpha/\partial\alpha = (1-2\alpha)\Psi^{(1)} + 2\alpha(\Psi^{(1)} + \Psi^{(2)}) > 0$ for $0 \le \alpha \le \frac{1}{2}$, and $\Psi_a(r) \le \Psi_{1/2}(r)$. Let $\Theta = \frac{2}{3}\mu^2\big(1 + \frac{j-2}{2\mu}\big) = \mu^2 - \frac{\mu r}{3} - \frac{\mu}{2}$. A direct expansion gives

$$\Theta - \Psi_{1/2}(r) = \frac{\mu^2}{6} + \frac{\mu r - r^2}{12} - \frac{r}{4} - \frac{1}{24} - \frac{\mu/16 - 1/8}{r} - \frac{3r - 2\mu}{64r^2(r+1)}.$$

Using $1 \le r \le \mu$ and $\mu > 2$, this is at least $\frac{\mu^2}{6} - \frac{5\mu}{16} + \frac{23}{384}$, which is positive for $\mu \ge \frac{5}{2}$. If $\Psi_a(r) < 0$ then $q_j > 1$. Otherwise (5.2) and Bernoulli's inequality $s^{-(j-2)} = (1 + \frac{1}{2\mu})^{j-2} \ge 1 + \frac{j-2}{2\mu}$ give

$$1 - q_j \le \frac{s^{j-2}\Psi_{1/2}(r)}{\mu^2} < \frac{s^{j-2}\,\Theta}{\mu^2} = \frac{2}{3}\,s^{j-2}\Big(1 + \frac{j-2}{2\mu}\Big) \le \frac{2}{3}.$$

The bound on δ_j. By (5.3), (5.5) and Lemma 5.7, using $\nu = \frac{a}{4}((1-a)\mu - 2a) > 0$ (as $\mu > 2 \ge \frac{2a}{1-a}$), $s \le 1$ and $j \ge 2$,

$$\begin{aligned}\delta_j &\le \frac{s^{j-2}}{\mu^2}\Big(\frac{a\mu(3\mu+1)}{2\mu(2\mu+1)} + \frac{a^2\mu}{8}\big(s\varrho(r-1) + s^{-1}\varrho(r+1)\big)\Big)\\ &\le \frac{a(3\mu+1)}{2\mu^2(2\mu+1)} + \frac{a^2}{8\mu}\big(\varrho(r-1) + s^{-1}\varrho(r+1)\big).\end{aligned}$$

The function $r/(r-1)^2$ is decreasing for $r > 1$, so $r \ge 1 + \frac{a}{2}$ gives $a^2r^2\varrho(r-1) = \frac{a^2 r}{(r-1)^2} \le 4 + 2a \le 5$. Moreover $s^{-1} = 1 + \frac{1}{2\mu} \le \frac{6}{5}$ and $a^2r^2\varrho(r+1) \le$

$\frac{a^2}{r+2} \le \frac{1}{12}$, so $a^2r^2s^{-1}\varrho(r+1) \le \frac{1}{10}$. Therefore, using $a \le \frac{1}{2}$,

$$3\delta_j \le \frac{3(3\mu+1)}{4\mu^2(2\mu+1)} + \frac{153}{80\,\mu\,r^2}.$$

Now $\frac{3(3\mu+1)}{4\mu^2(2\mu+1)} < \frac{1}{\mu+1/2} \le \frac{1}{j+1}$, because $8\mu^2 - 9\mu - 3 > 0$ for $\mu \ge \frac{5}{2}$ and $j+1 \le \mu + \frac{1}{2}$ by (5.6); and $\frac{153}{80\mu} \le \frac{153}{200} < 1$. □

Proposition 5.9. *Let $n \ge 4$, so that $j = 1$ is an internal index, and let $r = r_1 = \mu - \frac{1}{2}$.*

(a) *If $m \ge 3$, then $q_1 > \frac{1}{2}$ and $2\delta_1 < \frac{1}{2} + \frac{1}{r^2}$.*
(b) *If $m = 2$, then $q_0 < \frac{1}{2}$, $q_1 > \frac{4}{9}$ and $q_2 < \frac{3}{4}$.*

Proof. (a) Here $0 < a \le \frac{1}{3}$, $\mu \ge \frac{3}{2}$ and $r \ge 1 + \frac{a}{2} > 1$. By Lemma 5.6(b), $q_1 > 1 - 2a + \frac{3}{2}a^2 \ge \frac{1}{2}$. By (5.5) with $j = 1$, (5.3) and Lemma 5.7, noting $\nu > 0$ because $\mu > 1 \ge \frac{2a}{1-a}$,

$$\begin{aligned} 2\delta_1 &\le \frac{2s^{-1}}{\mu^2} \cdot \frac{a\mu(3\mu+1)}{2\mu(2\mu+1)} + \frac{a^2}{4\mu}\big(\varrho(r-1) + s^{-2}\varrho(r+1)\big) \\ &= \frac{a(3\mu+1)}{2\mu^3} + \frac{a^2}{4\mu}\big(\varrho(r-1) + s^{-2}\varrho(r+1)\big). \end{aligned}$$

The first term is at most $\frac{3\mu+1}{6\mu^3} \le \frac{22}{81}$, since this function decreases in μ. As before $a^2r^2\varrho(r-1) \le 4 + 2a \le \frac{14}{3}$, and $s^{-2} \le \frac{16}{9}$, $a^2r^2\varrho(r+1) \le \frac{a^2}{r+2} \le \frac{1}{27}$. Hence the second term is at most $\frac{1}{4\mu r^2}\big(\frac{14}{3} + \frac{16}{243}\big) \le \frac{575}{729\,r^2}$. Thus $2\delta_1 \le \frac{22}{81} + \frac{575}{729r^2} < \frac{1}{2} + \frac{1}{r^2}$.

(b) Here $a = \frac{1}{2}$, $\mu = \frac{2n-1}{4} \ge \frac{7}{4}$ and $\varsigma = \mu - \frac{1}{2}$. By Lemma 5.6, $q_0 = \frac{n-1}{2n-1} < \frac{1}{2}$ and $q_1 = \frac{4\varsigma^2 - \varsigma + 1}{8\varsigma^2}$; the inequality $q_1 > \frac{4}{9}$ is equivalent to $4\varsigma^2 - 9\varsigma + 9 > 0$, whose discriminant is negative. For q_2 we use (5.2) with $j = 2$, that is $1 - q_2 = \Psi(r_2)/\mu^2$, where $r_2 = \mu - \frac{3}{2} \ge \frac{1}{4}$. A direct expansion of (5.1) at $a = \frac{1}{2}$ gives

$$C(r_2) - \frac{r_2\mu^2}{4} = f(\mu) - \frac{\mu}{32r_2} + \frac{1}{16(r_2+1)}, \qquad f(\mu) = \frac{\mu^3}{4} - \frac{\mu^2}{4} - \frac{13\mu}{48} + \frac{1}{8}.$$

Since $\mu/r_2 = 1 + \frac{3}{2r_2} \le 7$, the right side exceeds $f(\mu) - \frac{7}{32}$. The function f is increasing for $\mu \ge \frac{7}{4}$ (its derivative is $\frac{3}{4}\mu^2 - \frac{\mu}{2} - \frac{13}{48} > 0$ there), and $f(\frac{7}{4}) - \frac{7}{32} = \frac{5}{768} > 0$. Hence $C(r_2) > \frac{r_2\mu^2}{4}$, that is, $q_2 < \frac{3}{4}$. □

5.6. **Arbitrary sets of joints.**

Proposition 5.10. *Let $n \ge 2$, $X \subseteq [n]$, $k = |X|$ and $\vartheta = 1 - \frac{k}{n}$. Then*

$$[t^p]\, Y_{\mathsf{Q}_n(G;X)} = n!\, m^n \sum_{j=0}^{\min(p,n-p)} W_j(\vartheta)\, d_{p-j}\, d_{n-p-j},$$

where $W_j(\vartheta) = U_j\big(1 - \vartheta + \vartheta q_j\big)$*, and the weights* $W_0(\vartheta), \dots, W_J(\vartheta)$ *are positive and satisfy* $W_j(\vartheta)^2 > W_{j-1}(\vartheta)W_{j+1}(\vartheta)$ *for* $1 \le j \le J-1$.

Proof. The formula follows from Propositions 5.1, 4.5, 5.4 and 5.5. Put $q_j(\vartheta) = 1 - \vartheta + \vartheta q_j$; it is positive by Lemma 5.6, and its second difference is $\vartheta\delta_j$. Fix $1 \le j \le J-1$. By Lemma 4.6 it suffices to show that

$$\log \frac{q_{j-1}(\vartheta)q_{j+1}(\vartheta)}{q_j(\vartheta)^2} < \frac{1}{j+1} + \frac{1}{r_j^2}. \tag{5.7}$$

If $\vartheta\delta_j \le 0$, the left side is at most 0 by (2.7). If $\vartheta\delta_j > 0$, then (2.7) and $q_j(\vartheta) \ge \vartheta q_j$ give

$$\log \frac{q_{j-1}(\vartheta)q_{j+1}(\vartheta)}{q_j(\vartheta)^2} < \frac{\vartheta\delta_j}{q_j(\vartheta)} \le \frac{\delta_j}{q_j}.$$

For $j \ge 2$, Proposition 5.8 gives $\delta_j/q_j < 3\delta_j < \frac{1}{j+1} + \frac{1}{r_j^2}$. For $j = 1$ and $m \ge 3$, Proposition 5.9(a) gives $\delta_1/q_1 < 2\delta_1 < \frac{1}{2} + \frac{1}{r_1^2}$. This proves (5.7) in these cases.

It remains to treat $j = 1$ for $m = 2$. By Proposition 5.9(b), $q_0(\vartheta) < 1 - \frac{\vartheta}{2}$, $q_2(\vartheta) < 1 - \frac{\vartheta}{4}$ and $q_1(\vartheta) > 1 - \frac{5\vartheta}{9} > 0$. For $0 \le \vartheta \le 1$,

$$2\Big(1 - \frac{5\vartheta}{9}\Big)^2 - \Big(1 - \frac{\vartheta}{2}\Big)\Big(1 - \frac{\vartheta}{4}\Big) = 1 - \frac{53\vartheta}{36} + \frac{319\vartheta^2}{648} \ge \frac{13}{648},$$

because this quadratic polynomial is decreasing on $[0,1]$ and takes the value $\frac{13}{648}$ at $\vartheta = 1$. Since $U_1^2/(U_0U_2) = \frac{2r_1^2}{r_1^2-1} > 2$ by Lemma 4.6, we obtain

$$W_1(\vartheta)^2 > 2U_0U_2\, q_1(\vartheta)^2 > U_0U_2\, q_0(\vartheta)\, q_2(\vartheta) = W_0(\vartheta)W_2(\vartheta).$$

□

Theorem 5.11. *Let* G *be a finite group of order* $m \ge 2$*,* $n \ge 2$ *and* $X \subseteq [n]$*. Then* $Y_{\mathsf{Q}_n(G;X)}$ *has degree* n*, positive palindromic coefficients, and its coefficients are strictly log-concave.*

Proof. Combine Proposition 5.10 with Lemma 2.8, where $\beta = a$. □

6. The trivial group

In this section $m = 1$, $a = 1$, and $Y_j = Y_{\mathsf{Q}_j(1)} = Y_{\mathsf{K}_{j+1}}$. Thus $d_l = c_l(2) = l + 1$ and $\tau_h = (2h-3)!!\,(2h-1)^{h-2}$. For $n \ge 1$ and $X \subseteq [n]$ with $|X| = k$ write $H_{n,k} = Y_{\mathsf{Q}_n(1;X)}$; by Corollary 3.3 and Lemma 2.2, $H_{n,0} = Y_{n-1}$, and $H_{n,1} = (1+t)Y_{n-1}$ because $\mathsf{Q}_n(1;\{i\})$ is the direct sum of K_n and the single coloop h_i.

6.1. **A positive decomposition.**

Lemma 6.1. *For $n \geq 2$ let $\Delta_n = (1+t)Y_{n-1} - \sum_{h\geq 1}\binom{n-1}{2h-1}\tau_h t^h Y_{n-2h}$. Then*

$$Y_n - n\Delta_n = \sum_{h\geq 1}\binom{n}{2h}\tau_h\, t^h\, Y_{n-2h}.$$

Proof. Let $\mathcal{F}(x) = \sum_n Y_n x^n/n!$, let v be defined by $ve^{-tv^2/2} = x$, and put $y = tv^2$. By Theorem 1.4, $\mathcal{F} = e^{y/2}(1-v)^{-1}(1-tv)^{-1}$, and by Lemma 5.2, $T(x) = t(v - tv^3/3)$. Let $\widetilde{T}(x) = \sum_{h\geq 1}\tau_h t^h x^{2h}/(2h)!$, so that $\widetilde{T}' = T$ and $\widetilde{T}(0) = 0$. We claim that

$$\widetilde{T}(x) = 1 - \Big(1 + \frac{y^2}{3}\Big)e^{-y/2}.$$

Both sides vanish at $v = 0$. Since $\frac{dx}{dv} = (1-y)e^{-y/2}$ and $\frac{dy}{dv} = 2tv$, the derivative of the right side with respect to v is

$$tv\, e^{-y/2}\Big(1 - \frac{4}{3}y + \frac{1}{3}y^2\Big) = tv\Big(1 - \frac{y}{3}\Big)(1-y)e^{-y/2} = T\,\frac{dx}{dv},$$

which proves the claim. Consequently

$$1 - (1+t)x + xT(x) - \widetilde{T}(x) = e^{-y/2}\big(1 + y - (1+t)v\big) = \mathcal{F}(x)^{-1},$$

since $1 + y - (1+t)v = (1-v)(1-tv)$. Taking the coefficient of $x^n/n!$, $n \geq 2$, in $\mathcal{F}(x)\big(1 - (1+t)x + xT(x) - \widetilde{T}(x)\big) = 1$ gives the lemma, as in the proof of Proposition 5.3. □

Proposition 6.2. *Let $n \geq 2$, $1 \leq k \leq n$ and $\ell = n - k$. Then*

$$H_{n,k} = \frac{k}{n}\,Y_n + \sum_{h=1}^{\lfloor n/2\rfloor}\Big(\frac{\ell}{n}\binom{n}{2h} - \binom{\ell}{2h}\Big)\tau_h\, t^h\, Y_{n-2h},$$

and all coefficients $\frac{\ell}{n}\binom{n}{2h} - \binom{\ell}{2h}$ are nonnegative.

Proof. For $2 \leq k' \leq n$, Proposition 3.6 gives

$$H_{n,k'} - H_{n,k'-1} = \Delta_n + \sum_h \binom{n-k'}{2h-1}\tau_h t^h Y_{n-2h}.$$

Summing over $k < k' \leq n$, using $H_{n,n} = Y_n$ and $\sum_{u=0}^{\ell-1}\binom{u}{2h-1} = \binom{\ell}{2h}$, we get $H_{n,k} = Y_n - \ell\Delta_n - \sum_h \binom{\ell}{2h}\tau_h t^h Y_{n-2h}$ for $k \geq 1$. Substituting Δ_n from Lemma 6.1 gives the formula. Nonnegativity follows from $\binom{\ell}{z}/\binom{n}{z} = \prod_{i=0}^{z-1}\frac{\ell-i}{n-i} \leq \frac{\ell}{n}$ for $z \geq 1$. □

For $k = 1$ the right side equals $(1+t)Y_{n-1}$, in accordance with Lemma 2.2; note that the argument above never deletes the last joint, which is a coloop.

6.2. **The weights.** For the rest of this section fix $n \ge 2$ and put

$$\lambda = \frac{n+1}{2}, \quad \mu = \frac{n}{2}, \quad s = \frac{\mu}{\lambda}, \quad J = \Big\lfloor \frac{n}{2} \Big\rfloor, \quad r_j = \lambda - j, \quad U_j = \frac{(\lambda - j)\lambda^{j-2}}{2 \cdot j!},$$

in accordance with (4.2). For $h \ge 1$ let $\kappa_h = \tau_h/(2h)! = \frac{(2h-1)^{h-3}}{2^h\, h!}$ and, for $0 \le j \le J$,

$$\omega_h(j) = \frac{(j)_{\downarrow h}\,(\lambda - h)^{j-h-2}}{\lambda^{j-2}} \quad (h \le j), \qquad \omega_h(j) = 0 \quad (h > j),$$

where $(j)_{\downarrow h} = j(j-1)\cdots(j-h+1)$. Note that $\lambda - h \ge \lambda - J \ge \frac{1}{2}$.

Proposition 6.3. *Let $1 \le k \le n$ and $\ell = n - k$. Then*

$$[t^p]\, H_{n,k} = n! \sum_{j=0}^{\min(p,n-p)} U_j\, q_j^{(k)}\, d_{p-j}\, d_{n-p-j},$$

$$q_j^{(k)} = \frac{k}{n} + \sum_{h=1}^{j} \Big(\frac{\ell}{n} - \frac{(\ell)_{\downarrow 2h}}{(n)_{\downarrow 2h}} \Big) \kappa_h\, \omega_h(j).$$

In particular $q_j^{(k)} \ge \frac{k}{n} > 0$.

Proof. Apply Proposition 4.5 with $m = 1$ to Y_{n-2h}, whose parameter is $\lambda - h$. After the shift $j \mapsto j - h$, the coefficient of t^p in $t^h Y_{n-2h}$ equals

$$(n-2h)! \sum_{j} \frac{(\lambda - j)(\lambda - h)^{j-h-2}}{2\,(j-h)!}\, d_{p-j}\, d_{n-p-j},$$

the sum running over $h \le j \le \min(p, n-p)$. Insert this into Proposition 6.2, divide by $n!$, and use $\binom{n}{2h}\frac{(n-2h)!}{n!} = \frac{1}{(2h)!}$ and $\binom{\ell}{2h}\frac{(n-2h)!}{n!} = \frac{(\ell)_{\downarrow 2h}}{(2h)!\,(n)_{\downarrow 2h}}$. Dividing the resulting weight by U_j gives $q_j^{(k)}$. □

6.3. **Monotonicity of σ_h.** Put $\sigma_0 = 1$ and, for $1 \le h \le J$,

$$\sigma_h = \frac{1}{k(2h-1)} \Big(\ell - \frac{n\,(\ell)_{\downarrow 2h}}{(n)_{\downarrow 2h}} \Big),$$

so that

$$q_j^{(k)} = \frac{k}{n} \Big(1 + \sum_{h=1}^{j} \sigma_h (2h-1) \kappa_h\, \omega_h(j) \Big). \tag{6.1}$$

For $k = 1$ we have $(n-1)_{\downarrow 2h}/(n)_{\downarrow 2h} = (n-2h)/n$ and hence $\sigma_h = 1$ for $0 \le h \le J$. We write $Q_j = q_j^{(1)} = \frac{1}{n}\big(1 + \sum_{h=1}^{j}(2h-1)\kappa_h\, \omega_h(j)\big)$.

Lemma 6.4. *For every $1 \le k \le n$, the numbers σ_h, $0 \le h \le J$, satisfy $1 = \sigma_0 \ge \sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_J \ge 0$.*

Proof. If $k = n$ then $\sigma_h = 0$ for $1 \le h \le J$. Let $k < n$, so $\ell \ge 1$. For $0 \le z \le n-1$ let $\Pi_z = 1 - \binom{\ell-1}{z}/\binom{n-1}{z}$; this is the probability that a uniformly random z-subset of a set with k marked and $\ell-1$ unmarked elements contains a marked element. Since $(\ell)_{\downarrow 2h}/(n)_{\downarrow 2h} = \frac{\ell}{n}\binom{\ell-1}{2h-1}/\binom{n-1}{2h-1}$, we have $\sigma_h = \ell\,\Pi_{2h-1}/(k(2h-1))$. Let $\Delta\Pi_z = \Pi_{z+1} - \Pi_z = \frac{k}{n-1-z}\binom{\ell-1}{z}/\binom{n-1}{z} \ge 0$. If $\Delta\Pi_z > 0$ and $z+1 \le n-2$, then $\Delta\Pi_{z+1}/\Delta\Pi_z = \frac{\ell-1-z}{n-2-z} \le 1$; and once $\Delta\Pi_z = 0$, all later increments vanish. Thus Π is concave on $\{0,\dots,n-1\}$ with $\Pi_0 = 0$, so Π_z/z is nonincreasing for $z \ge 1$. Hence σ_h is nonincreasing for $h \ge 1$, and $\sigma_h \ge 0$. Finally $\sigma_1 = \ell\,\Pi_1/k = \frac{\ell}{n-1} \le 1$. □

Lemma 6.5. *Let $1 \le j \le J-1$, $r = r_j$, and for $1 \le h \le j$ put $\epsilon_h = \big(\omega_h(j-1) + \omega_h(j+1) - 2\omega_h(j)\big)/\omega_h(j)$. Then*

$$\epsilon_h = \frac{h\,\big(h(j+r^2) - \lambda^2\big)}{\lambda\, j\,(\lambda-h)(j+1-h)},$$

and the sequence $\max(\epsilon_h, 0)$ is nondecreasing in h and strictly increasing once positive. Moreover $\omega_{j+1}(j-1) = \omega_{j+1}(j) = 0 < \omega_{j+1}(j+1)$, and ω_h vanishes at $j-1, j, j+1$ for $h \ge j+2$.

Proof. For $1 \le h \le j$ we have $\omega_h(j) > 0$,

$$\frac{\omega_h(j+1)}{\omega_h(j)} - 1 = \frac{h(r-1)}{\lambda(j+1-h)}, \qquad \frac{\omega_h(j-1)}{\omega_h(j)} - 1 = -\frac{hr}{j(\lambda-h)},$$

where for $h = j$ the second identity correctly gives $\omega_j(j-1) = 0$. Adding the two and using $\lambda = r + j$ gives the formula for ϵ_h. In it, the denominator is positive and decreasing in h, and once $h(j+r^2) - \lambda^2 > 0$ the numerator is positive and increasing in h. The last statements follow from the definition of ω_h. □

Proposition 6.6. *Let $1 \le j \le J-1$, $1 \le k \le n$, and write $\delta q_j = q_{j-1} + q_{j+1} - 2q_j$ for a sequence q. Then*

$$\frac{\delta q_j^{(k)}}{q_j^{(k)}} \le \max\Big(0, \frac{\delta Q_j}{Q_j}\Big).$$

Proof. Let $L = \max(0, \delta Q_j/Q_j)$. By (6.1), $\frac{n}{k}\big(\delta q_j^{(k)} - L q_j^{(k)}\big) = \sum_{h=0}^{j+1} \sigma_h x_h$, where

$$x_0 = -L, \qquad x_h = (2h-1)\kappa_h\big(\omega_h(j-1) + \omega_h(j+1) - (2+L)\omega_h(j)\big) \quad (h \ge 1),$$

and $\sum_h x_h = n(\delta Q_j - L Q_j) \le 0$. Here σ_{j+1} is defined by the same formula if $j+1 \le J$; since $j \le J-1$, this is the case. For $1 \le h \le j$ the sign of x_h is that of $\epsilon_h - L$. By Lemma 6.5 and $L \ge 0$, the sequence $x_0, x_1, \dots, x_j$ is nonpositive up to some index and nonnegative afterwards, and $x_{j+1} > 0$. Let h_0 be the first index with $x_{h_0} > 0$ and $\sigma_* = \sigma_{h_0}$. By Lemma 6.4, $\sigma_h \ge \sigma_*$ for $h < h_0$, where $x_h \le 0$, and $\sigma_h \le \sigma_*$ for $h \ge h_0$, where $x_h \ge 0$. Hence $\sum_h \sigma_h x_h \le \sigma_* \sum_h x_h \le 0$, that is, $\delta q_j^{(k)} \le L q_j^{(k)}$. □

6.4. **The case of one joint.**

Proposition 6.7. *For $0 \le j \le J$ we have $Q_j = s^{j-2}\,\Omega(r_j)/n$, where*

$$\Omega(r) = 2 - \frac{r+\frac12}{\mu} + \frac{3}{8\mu}\Big(\frac1r + \frac{1}{r+1}\Big) + \frac{1}{4r^2(r+1)}.$$

Proof. By Lemma 2.2, $H_{n,1} = (1+t)Y_{n-1}$. By (4.1) for Y_{n-1} with $m=1$ (parameter $\frac{n}{2} = \mu$) and (2.5), the coefficient of t^p in $H_{n,1}/n!$ is $\frac1n[\xi^{n-p}\eta^p]\,(\xi+\eta)K_1(\mu)$, where $K_\beta(\theta) = (1-z)e^{\theta z}R^{-\beta}$. For a series $\Phi \in \mathbb{R}[[\xi,\eta]]$ we call the coefficient sequence $\big([\xi^{n-p}\eta^p]\,\Phi\big)_{0\le p\le n}$ of its homogeneous component of degree n the *row* of Φ. Write $\xi+\eta = 1+z-R$. The component of degree n of $RK_1(\mu) = (1-z)e^{\mu z}$ vanishes: it is zero for n odd, and for n even it equals $w_{n/2}(\mu)z^{n/2} = 0$ because $\mu = n/2$. Hence the row is that of $\frac1n(1-z^2)e^{\mu z}R^{-1}$. Corollary 2.5 with $\beta = 1$ gives the weights $\frac{1}{2n}\big(\frac{v'_j}{r_j} - \frac{v'_{j-1}}{r_j+1}\big)$ with $v'_j = [z^j](1-z^2)e^{\mu z} = \frac{\mu^{j-2}}{j!}\big(\mu^2 - j(j-1)\big)$. By (2.4) the rows $[z^jR^{-2}]_n$, $0\le j\le J$, are linearly independent: the coefficient of $\xi^{n-p}\eta^p$ vanishes for $p<j$ and equals $d_{n-2j}d_0 \ne 0$ for $p=j$. Hence these weights coincide with U_jQ_j from Proposition 6.3. Dividing by U_j and substituting $j = \mu + \frac12 - r$, we obtain $nQ_j = s^{j-2}\,\Omega_1(r_j)$ with

$$\Omega_1(r) = \frac{\mu^2 - j(j-1)}{r^2} - \frac{j\,\big(\mu^2 - (j-1)(j-2)\big)}{\mu\, r(r+1)},$$

and a direct expansion shows $\Omega_1 = \Omega$. □

Proposition 6.8. *Let $1 \le j \le J-1$ and $r = r_j$. Then $\dfrac{\delta Q_j}{Q_j} < \dfrac{1}{j+1} + \dfrac{1}{r^2}$.*

Proof. Since $J \ge 2$, we have $n \ge 4$, $\mu \ge 2$, and $\frac32 \le r \le \mu - \frac12$; moreover r is an integer or a half-integer. By Proposition 6.7,

$$\frac{\delta Q_j}{Q_j} = \frac{\mathcal{L}_s\Omega(r)}{\Omega(r)},$$

where $\mathcal{L}_s f(r) = s\,f(r-1) + s^{-1}f(r+1) - 2f(r)$ is the operator of Section 5, now with $s = \mu/\lambda$ as defined in this section. Write $\Omega = \Lambda + \Xi$ with $\Lambda(r) = 2 - (r+\frac12)/\mu \ge 1$ and $\Xi(r) = \frac{3}{8\mu}(\frac1r + \frac{1}{r+1}) + \frac14\varrho(r)$, where $\varrho(r) = 1/(r^2(r+1))$ as before. Put $\varepsilon = s + s^{-1} - 2 = \frac{1}{4\mu\lambda}$. Since Λ is affine with slope $-1/\mu$ and $s - s^{-1} = -\frac{4\mu+1}{2\mu(2\mu+1)}$, a direct computation gives $\mathcal{L}_s\Lambda = -\varepsilon\big(2 + (r+\frac32)/\mu\big)$. For a positive decreasing function f,

$$\mathcal{L}_s f(r) - \nabla^2 f(r) - \varepsilon f(r) = (s-1)\big(f(r-1) - f(r)\big) + (s^{-1}-1)\big(f(r+1) - f(r)\big),$$

which is nonpositive; here $\nabla^2 f(r) = f(r-1) + f(r+1) - 2f(r)$. The function Ξ is positive and decreasing for $r > 0$, and $\Xi(r) \le \frac{2}{5\mu} + \frac{2}{45} < 2$ because $r \ge \frac32$. Therefore $\mathcal{L}_s\Omega < \nabla^2\Xi$, and since $\Omega \ge \Lambda \ge 1$, it suffices to show

$$\nabla^2\Xi(r) < \Big(\frac{1}{j+1} + \frac{1}{r^2}\Big)\Lambda(r) \quad \text{whenever } \nabla^2\Xi(r) > 0. \tag{6.2}$$

We have $\nabla^2 \frac{1}{r} = \frac{2}{r(r^2-1)}$ and $\nabla^2 \frac{1}{r+1} = \frac{2}{r(r+1)(r+2)}$, so

$$\nabla^2\Big(\Xi - \frac{\varrho}{4}\Big) = \frac{3}{4\mu}\Big(\frac{1}{r(r^2-1)} + \frac{1}{r(r+1)(r+2)}\Big) \le \frac{16}{35\mu} < \frac{1}{j+1},$$

where we used that the bracket decreases in r and equals $\frac{64}{105}$ at $r = \frac{3}{2}$, and that $j+1 = \mu + \frac{3}{2} - r \le \mu$. If $r \ge 2$, then

$$\begin{aligned} r^2\,\nabla^2 \frac{\varrho}{4} &= \frac{r}{4(r-1)^2} + \frac{r^2}{4(r+1)^2(r+2)} - \frac{1}{2(r+1)} \\ &\le \frac{r}{4(r-1)^2} + \frac{1}{4(r+2)} \le \frac{9}{16}, \end{aligned}$$

since both terms decrease in r; this gives (6.2) with $\Lambda \ge 1$. If $r = \frac{3}{2}$ and $j \ge 2$, then $\mu = j+1 \ge 3$, $\Lambda = 2 - \frac{2}{\mu} \ge \frac{4}{3}$, and $r^2\nabla^2 \frac{\varrho}{4} = \frac{9}{4} \cdot \frac{928}{1575} = \frac{232}{175} < \frac{4}{3}$, which again gives (6.2). Finally, if $r = \frac{3}{2}$ and $j = 1$, then $\mu = 2$, $\Lambda = 1$, and $\nabla^2\Xi = \frac{8}{35} + \frac{928}{1575} = \frac{184}{225} < \frac{1}{2} + \frac{4}{9}$. □

Theorem 6.9. *Let $n \ge 2$ and $1 \le k \le n$. Then $H_{n,k}$ has degree n, positive palindromic coefficients, and its coefficients are strictly log-concave.*

Proof. By Proposition 6.3, the coefficients have the form of Lemma 2.8 with $\beta = 1$ and weights $U_j q_j^{(k)} > 0$. If $J \le 1$ there is nothing more to prove. Let $1 \le j \le J-1$. If $\delta q_j^{(k)} \le 0$, then $q_{j-1}^{(k)} q_{j+1}^{(k)} \le (q_j^{(k)})^2$ by (2.7). Otherwise (2.7), Propositions 6.6 and 6.8, and Lemma 4.6 give

$$\log \frac{q_{j-1}^{(k)} q_{j+1}^{(k)}}{(q_j^{(k)})^2} < \frac{\delta q_j^{(k)}}{q_j^{(k)}} \le \max\Big(0, \frac{\delta Q_j}{Q_j}\Big) < \frac{1}{j+1} + \frac{1}{r_j^2} \le \log \frac{U_j^2}{U_{j-1}U_{j+1}}.$$

In both cases the weights $U_j q_j^{(k)}$ are strictly log-concave at j, and Lemma 2.8 completes the proof. □

7. Proof of Theorem 1.2 and examples

Proof of Theorem 1.2. Part (i) is Corollary 3.3. If $n = 1$, the matroid is empty or consists of a single coloop, and $Y = 1$ or $1 + t$ by Lemma 2.2(d). Let $n \ge 2$. If $m \ge 2$, parts (ii) and (iii) follow from Proposition 5.1 and Theorem 5.11; all ingredients depend only on n, m and k. If $m = 1$ and $k \ge 1$, they follow from Proposition 6.2 and Theorem 6.9. If $m = 1$ and $k = 0$, then $\mathsf{Q}_n(1;\emptyset) \cong \mathsf{K}_n = \mathsf{Q}_{n-1}(1)$, and Theorem 4.7 applies when $n - 1 \ge 2$, while $Y_{\mathsf{K}_2} = 1 + t$.

For (iv), let first $m \geq 2$. By Proposition 4.5, $Y_n(0) = \prod_{i=0}^{n-1}(1+im)$. By Propositions 5.4 and 5.5, Lemma 5.6(a) and $U_0 = \frac{a}{n+a}$,

$$Y_n^\circ(0) = n!\, m^n U_0 q_0\, d_n = m^n \frac{a(1-a)(n-1)}{(n+a)(n+a-1)}\,(a+1)\cdots(a+n)$$
$$= (m-1)(n-1)\prod_{h=1}^{n-2}(mh+1).$$

By Proposition 5.1,

$$y_0 = \frac{k}{n}\big(1+(n-1)m\big)\prod_{h=1}^{n-2}(1+hm) + \frac{n-k}{n}(m-1)(n-1)\prod_{h=1}^{n-2}(mh+1),$$

and $k(1+(n-1)m) + (n-k)(m-1)(n-1) = n\big((m-1)(n-1)+k\big)$. If $m = 1$ and $k \geq 1$, only the term $\frac{k}{n}Y_n$ in Proposition 6.2 contributes to the constant coefficient, which equals $\frac{k}{n}\,n! = k(n-1)!$, in accordance with (iv). The equality $y_n = y_0$ is palindromicity. □

Proof of Corollary 1.3. The Dowling geometries are the case $X = [n]$, the braid matroid K_n is $\mathsf{Q}_{n-1}(1)$, the matroids of types B_n and D_n are $\mathsf{Q}_n(\mathbb{Z}/2\mathbb{Z}; X)$ with $X = [n]$ and $X = \emptyset$, and the graphs in the last family are those of Section 3 for $m = 1$. □

Example 7.1. As an independent check, we computed $Y_{\mathsf{Q}_n(G;X)}$ for the groups and sizes in Table 1 directly from the definition: the lattice of flats was generated from the rank function (3.1) by closure, and P, Z and Y were obtained from Theorem 2.1 and (2.1) with exact arithmetic. This computation uses neither the description of flats in Section 3 nor any generating function. In every case the result agrees with Theorem 1.2 and the formulas of Sections 4–6; we list only $y_0, \ldots, y_{\lfloor r/2 \rfloor}$. The symmetric group $\mathfrak{S}_3$ and the cyclic group $\mathbb{Z}/6\mathbb{Z}$ give the same polynomials, and for $m \geq 2$ the coefficients are affine functions of k, as predicted by Proposition 5.1.

TABLE 1. Inverse Z-polynomials computed from the lattice of flats.

G	n	k	$y_0, y_1, \ldots$	G	n	k	$y_0, y_1, \ldots$
1	4	0	6, 12	$\mathbb{Z}/2$	4	0	45, 108, 130
1	4	1	6, 18, 24	$\mathbb{Z}/2$	4	1	60, 141, 168
1	4	2	12, 34, 44	$\mathbb{Z}/2$	4	2	75, 174, 206
1	4	3	18, 48, 61	$\mathbb{Z}/2$	4	3	90, 207, 244
1	4	4	24, 60, 75	$\mathbb{Z}/2$	4	4	105, 240, 282
$\mathbb{Z}/3$	3	0	16, 27	$\mathbb{Z}/6$, $\mathfrak{S}_3$	3	0	70, 108
$\mathbb{Z}/3$	3	1	20, 34	$\mathbb{Z}/6$, $\mathfrak{S}_3$	3	1	77, 121
$\mathbb{Z}/3$	3	2	24, 41	$\mathbb{Z}/6$, $\mathfrak{S}_3$	3	2	84, 134
$\mathbb{Z}/3$	3	3	28, 48	$\mathbb{Z}/6$, $\mathfrak{S}_3$	3	3	91, 147

The same check was carried out with exact arithmetic for the weight representations of Propositions 4.5, 5.10 and 6.3 against the recursions of Propositions 5.1 and 6.2, for $2 \leq n \leq 24$, all admissible k, and $m \in \{1, 2, 3, 4, 6, 10, 100, 1000\}$. These computations are not used in the proofs.

USE OF AI-ASSISTED TOOLS

AI-assisted tools were used in the research and preparation of this manuscript.

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